Model Name: GA-H61M-DS2 DVI *Revision 1.0*

SHEET	TITLE
01	COVER SHEET
02	BOM & PCB MODIFY HISTORY
03	BLOCK DIAGRAM
04	CPU_LGA1155-A
05	CPU_LGA1155-B
06	CPU_LGA1155-C
07	DDR III CHANNEL A
08	DDR III CHANNEL B
09	PCH_FDI,DMI,USB,PCIE,NVRAM
10	PCH_DP,CLK BUFFER
11	PCH_HOST,SATA,PCI
12	PCH_GPIO,CTRL,AUDIO
13	PCH_PWR,GND
14	PCI EXPRESS*16 SLOT
15	PCI EXPRESS*1x2 SLOT
16	ITE 8728 LPC IO
17	COMA,KB_MS,R_USB,-PROCHOT
18	HWM,FAN CTRL,OV
19	DUAL BIOS
20	FP,FUSB,SPK,SATALED
21	ALC887
22	REAR AUDIO JACK
23	REALTEK 8111E/USB_LAN
24	DISCRETE POWER
25	ATX,-S_WARN,-S_ACK,5VDUAL
26	Richtek_RT8120_CPU_VTT
27	VCORE INTERSIL_95836_1

SHEET	TITLE
28	VCORE INTERSIL_95836_2
29	VCORE INTERSIL_95836_3
30	LPT

Gigabyte Confidential
Do not Copy

Gigabyte Technology

Title: Cover Sheet

Size: Custom | Document Number: GA-H61M-DS2 DVI | Rev: 1.0

Date: Tuesday, September 25, 2012 | Sheet 1 of 31

Model Name: GA-H61M-DS2 DVI

Revision 1.0

Component value change history

2012/09/17

Data	Change Item	Reason
2012/09/17	change from H61M-DS2 Rev 2.22	9M61MDS2D-00-01
2012/09/21 9M61MDS2D-00-10A	*** 更換 *** 新舊線路中,元件值不同;共 10 個********** DR447,'3.4K/4/1變更爲'3K/4/1 CBC12,CBC25,'0.1u/4/X7R/16V/K變更爲'10u/6/X5R/6.3V/M DR463,'4.12K/4/1變更爲'2.87K/4/1 DR448,'270K/4變更爲'243K/4/1 DR162,'487/4/1變更爲'510/4/1 DR47,'620/4/1變更爲'499/4/1 R96,R4993,'10K/4/1變更爲'8.06K/4/1 DR464,'470K/4/1變更爲'270K/4 *** 增加 *** 只在新線路中,不在舊線路中;共 7 個********** D4,'BAT54C/SOT23/200mA R391,'150K/4 R390,'270K/4 R4945,'8.2K/4 C135,'0.1u/4/X7R/16V/K Q56,'MMBT2222A/SOT23/600mA/40 R137,'8.2K/4/1 *** 刪除 *** 只在舊線路中,不在新線路中;共 2 個********** R4991,'0/4 R4980,'8.2K/4
2012/09/21 9M61MDS2D-00-10B	*** 增加 *** 只在新線路中,不在舊線路中;共 18 個********** BC41,BC45,BC47,BC52,BC53,BC54,BC55,BC60,BC61,BC62,BC63,BC64,BC65,BC66,BC67,BC70,BC72,BC74, '10u/6/X5R/6.3V/M

Circuit or PCB layout change

DATE	Change Item	Reason
2012/09/17	change from H61M-DS2 Rev 2.22	Rev 0.1
2012/09/21 Rev 1.0	1.Add R4994 0ohm 2.Change R38 Net VCC3 to 3VDUAL_PCH 3.RS1 移至DQ6旁。 4.CN1 轉90度，已Shift Net (KBCLK to pin1、KBDLK to pin3、MSCLK to pin7、MSDLK to pin5)	

Gigabyte Confidential
Do not Copy

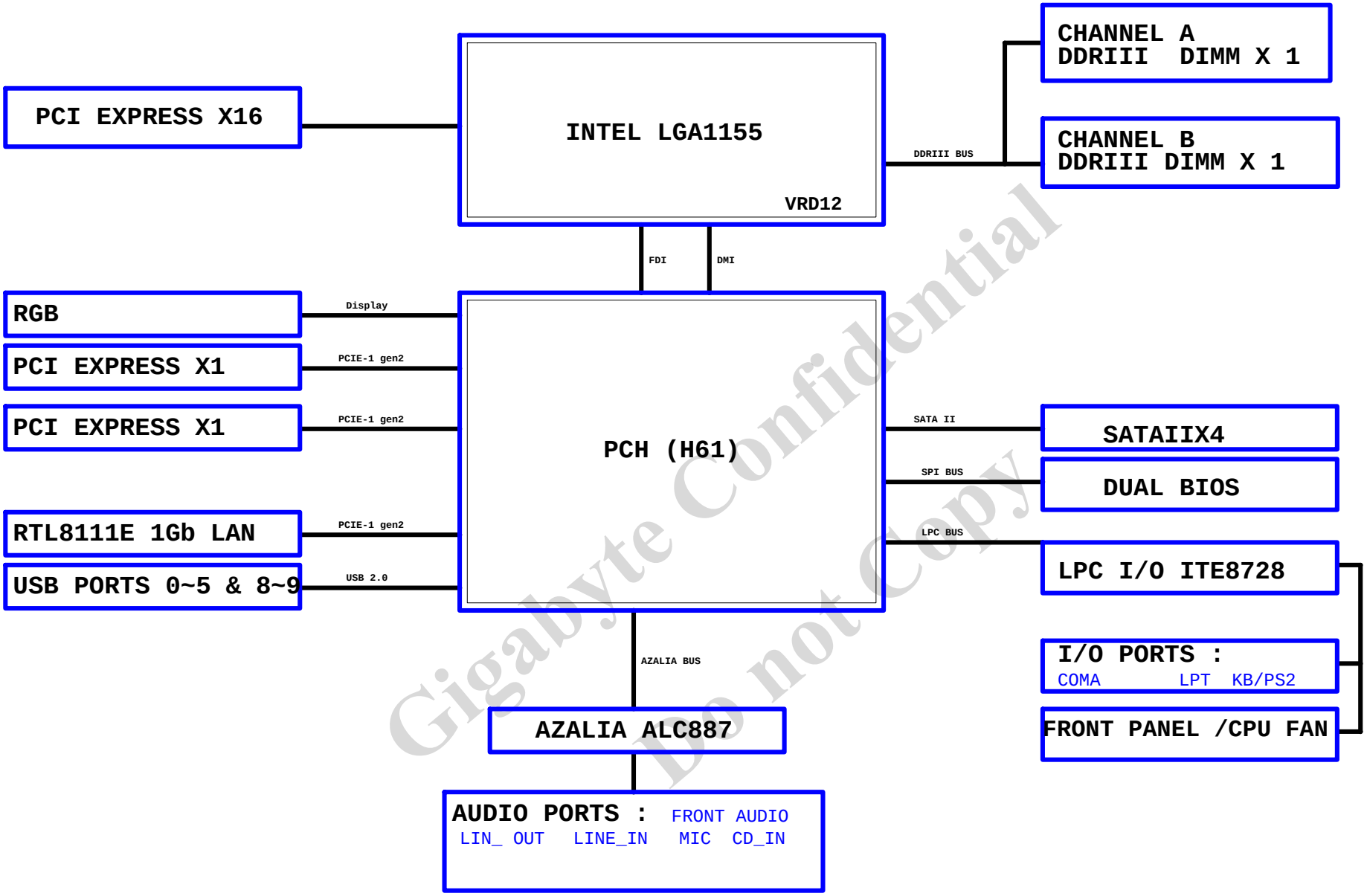
Gigabyte Technology

Title: BOM & PCB MODIFY HISTORY

Size: Custom | Document Number: GA-H61M-DS2 DVI | Rev: 1.0

Date: Tuesday, September 25, 2012 | Sheet 2 of 31

BLOCK DIAGRAM



Gigabyte Confidential
Do not Copy

www.schematic-x.blogspot.com

Gigabyte Technology		
Title: BLOCK DIAGRAM		
Size: C	Document Number: GA-H61M-DS2 DVI	Rev: 1.0
Date: Tuesday, September 25, 2012	Sheet 3 of 31	

CPUPWROK
R132 1K/4/1
C127 1n/4/X7R/50V/K

CPUE
BCLK[0] BCLK#[0] VIDSCLK VIDSOUT VIDALERT# UNCOREPWRGOOD SM_DRAMPWROK RESET# PM_SYNC PECI CATERR# PROCHOT# THERMTRIP# SKTOCC# FC_K32 SM_VREF CFG[0] CFG[1] CFG[2] CFG[3] CFG[4] CFG[5] CFG[6] CFG[7] CFG[8] CFG[9] CFG[10] CFG[11] CFG[12] CFG[13] CFG[14] CFG[15] CFG[16] CFG[17] RSVD_016 RSVD_023 RSVD_028 RSVD_029
VCCIO_SELECT VCCSA_VID_0 VCCSA_SENSE VCC_SENSE VSS_SENSE VCCIO_SENSE VSSIO_SENSE VCCAXG_SENSE VSSAXG_SENSE TDO TDI TCK TMS TRST# PRDY# PREQ# DBR# BCLK_ITP BCLK_ITP# BPM#[0] BPM#[1] BPM#[2] BPM#[3] BPM#[4] BPM#[5] BPM#[6] BPM#[7] RSVD_024 RSVD_030 RSVD_037 RSVD_036 RSVD_033 RSVD_040 RSVD_039 RSVD_018 RSVD_020 RSVD_038 RSVD_032 RSVD_034 RSVD_035 VCC_VALIDATION_SENSE VSS_VALIDATION_SENSE VCCAXG_VALIDATION_SENSE VSSGT_VALIDATION_SENSE
5 OF 10
LGA1155[10SC1-F01155-01R_10SC1-F01155-02R]

H61M-S2C-B3 remove

CFG	H	L	NOTE
0	RSVD	RSVD	RSVD
1	RSVD	RSVD	RSVD
2	**NORM**	**Reverse**	LANE REVERSAL[0],x16
3	RSVD	RSVD	RSVD
4	RSVD	RSVD	RSVD
7	RSVD	RSVD	RSVD
8	RSVD	RSVD	RSVD
9	RSVD	RSVD	RSVD
10	RSVD	RSVD	RSVD
11	RSVD	RSVD	RSVD
12	RSVD	RSVD	RSVD
13	RSVD	RSVD	RSVD
14	RSVD	RSVD	RSVD
15	RSVD	RSVD	RSVD
16	RSVD	RSVD	RSVD
17	RSVD	RSVD	RSVD

CFG6	CFG5	PCIE CONFIG
1	1	1X16 , Default
1	0	2X8
0	1	RSVD
0	0	X8,X4,X4

CFG 0-17 all internal PULL-UP

DRAM_PWROK
BC93 22p/4/NPO/50V/J/X

R199 8.2K/4/X
CPU_VTT
-THRMTRIP
R204 0/4/SHT/M/X
Q25 MMBT2222A/SOT23/600mA/40/X
-THRMTRIP_PCH -THRMTRIP_PCH 11,17
R196 8.2K/4/X
CPU_VTT
DIS_T
-THRMTRIP
S0T23

3VDUAL VCC3
R225 1K/4/1
R222 200/4/1
1.1V分壓
CPURST
R223 100/4/1
BC102 1n/4/X7R/50V/K
R224 8.2K/4
16 -PFMRST1
Q28 MMBT2222A/SOT23/600mA/40
Q29 MMBT2222A/SOT23/600mA/40
S0T23

CPUD
FDI_FSYNC0 FDI_LSYNC0 FDI_FSYNC1 FDI_LSYNC1 FDI_INT FDI_RCOMP FDI_TX[0] FDI_TX#[0] FDI_TX[1] FDI_TX#[1] FDI_TX[2] FDI_TX#[2] FDI_TX[3] FDI_TX#[3] FDI_TX[4] FDI_TX#[4] FDI_TX[5] FDI_TX#[5] FDI_TX[6] FDI_TX#[6] FDI_TX[7] FDI_TX#[7] FDI_COMPIO FDI_ICOMPO
FDI LINK
4 OF 10
LGA1155[10SC1-F01155-01R_10SC1-F01155-02R]
R170 24.9/4/1

FDI_TXP[0..7] FDI_TXN[0..7] EXP_A_TXP[0..15] EXP_A_TXN[0..15] EXP_A_RXP[0..15] EXP_A_RXN[0..15]

CPU_VTT
R89 90.9/4/1/X VIDSLCK
R90 100/4/1 VIDSOUT
R91 100/4/1 -VIDALRT
RN6 51/8P4R/4
TDI TDO TMS -HPRDY
RN19 51/8P4R/4
-THRMTRIP -PROCHOT TCK -TRST

DDR_15V
R193 100/4/1
SM_VREF
R191 100/4/1
C85 0.1u/4/X7R/16V/K

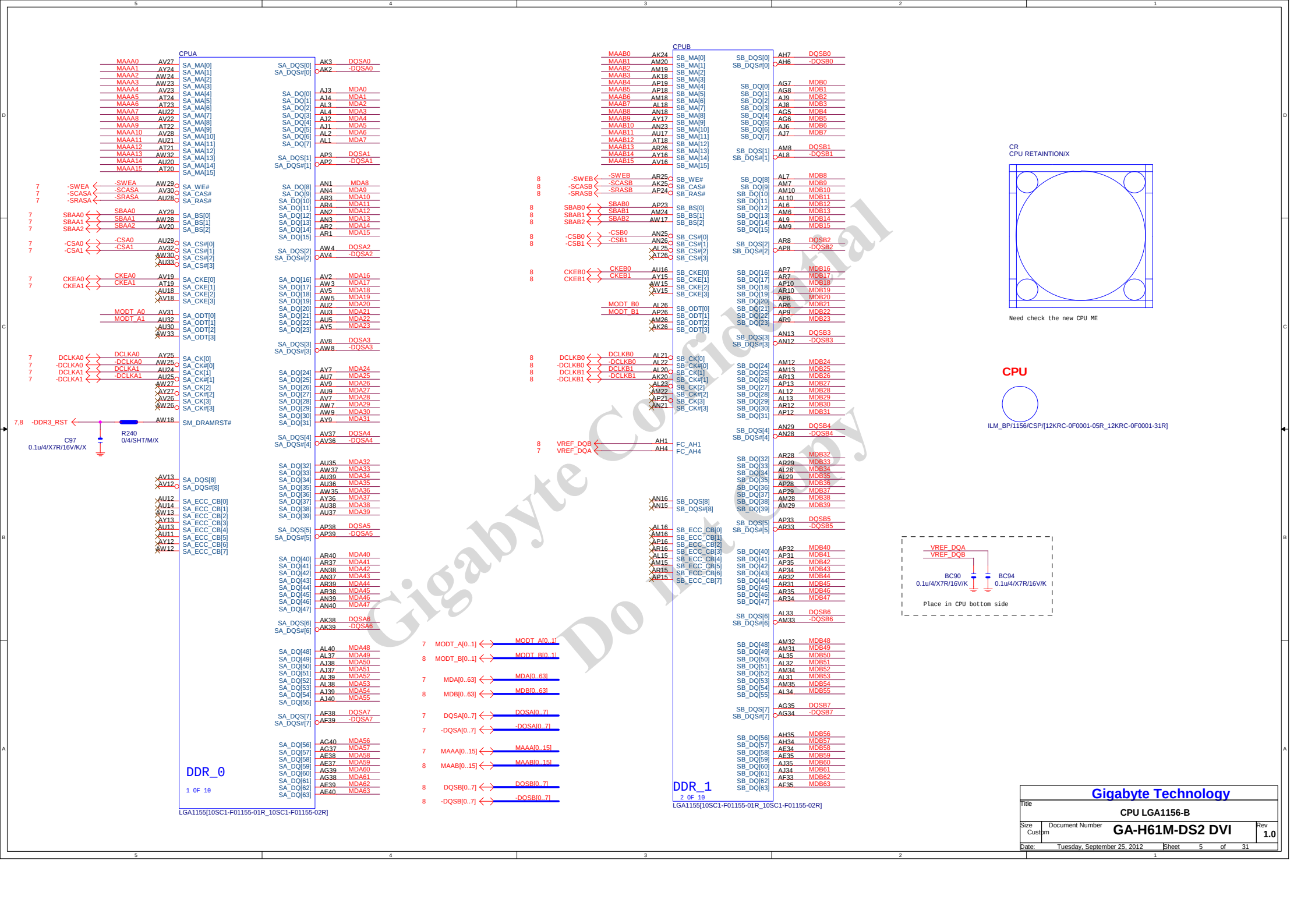
Stitching caps for PCIE,DMI,FDI bus
CPU_VTT PBC3 PBC9 PBC8 PBC6 PBC7 PBC13 PBC16 PBC15
0.1u/4/X7R/16V/K/X

CPU_VTT PBC10 PBC14 PBC12 PBC11 PBC4 PBC2 PBC5 PBC1
0.1u/4/X7R/16V/K/X

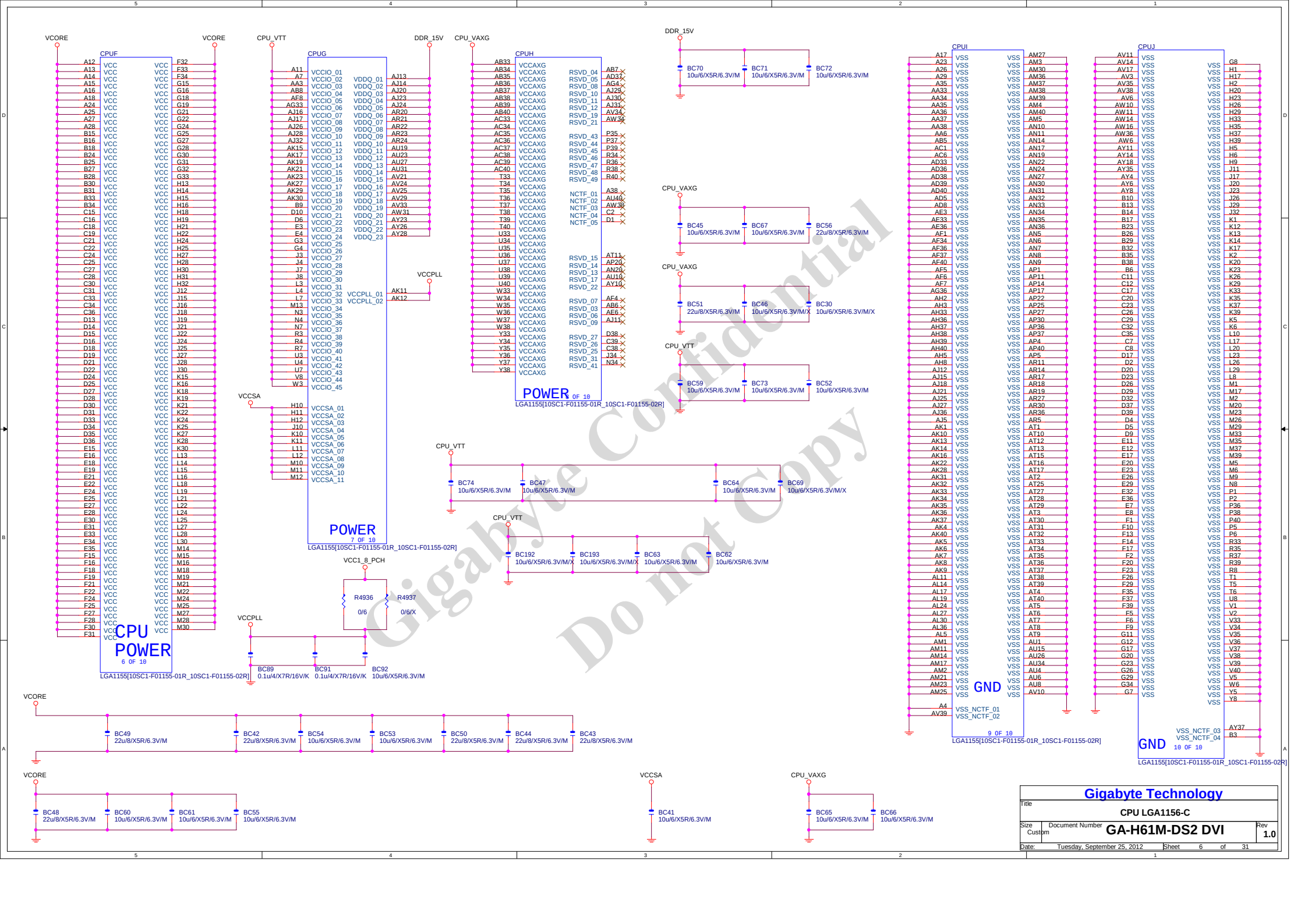
Stitching caps for PCIE,DMI,FDI bus

CPUC
PEG_RX[0..15] PEG_RX#[0..15] PEG_TX[0..15] PEG_TX#[0..15]
PEG
DMI_RX[0..3] DMI_RX#[0..3] DMI_TX[0..3] DMI_TX#[0..3]
DMI
PE_RX[0..3] PE_RX#[0..3] PE_TX[0..3] PE_TX#[0..3]
GEN
PEG_ICOMPO PEG_RCOMPO PEG_ICOMPI
3 OF 10
LGA1155[10SC1-F01155-01R_10SC1-F01155-02R]
B5 GRCOMP R48 24.9/4/1 CPU_VTT
C4
B4
W=12 mil out of CPU
S=15 mil out of CPU

Gigabyte Technology				
Title: CPU LGA1155-A				
Size: Custom	Document Number: GA-H61M-DS2 DVI			Rev: 1.0
Date: Tuesday, September 25, 2012		Sheet 4 of 31		

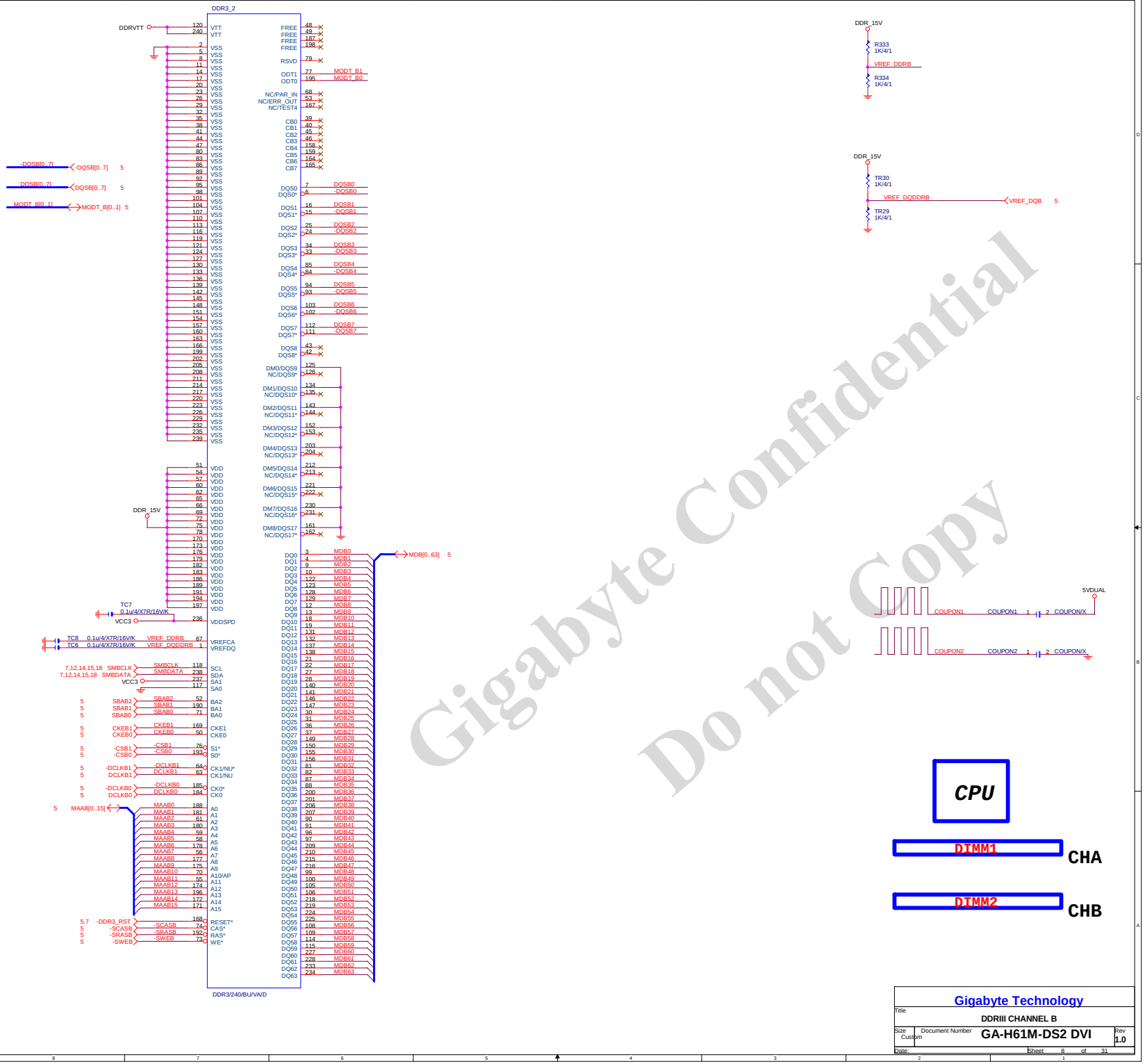


Gigabyte Confidential
Do Not Copy



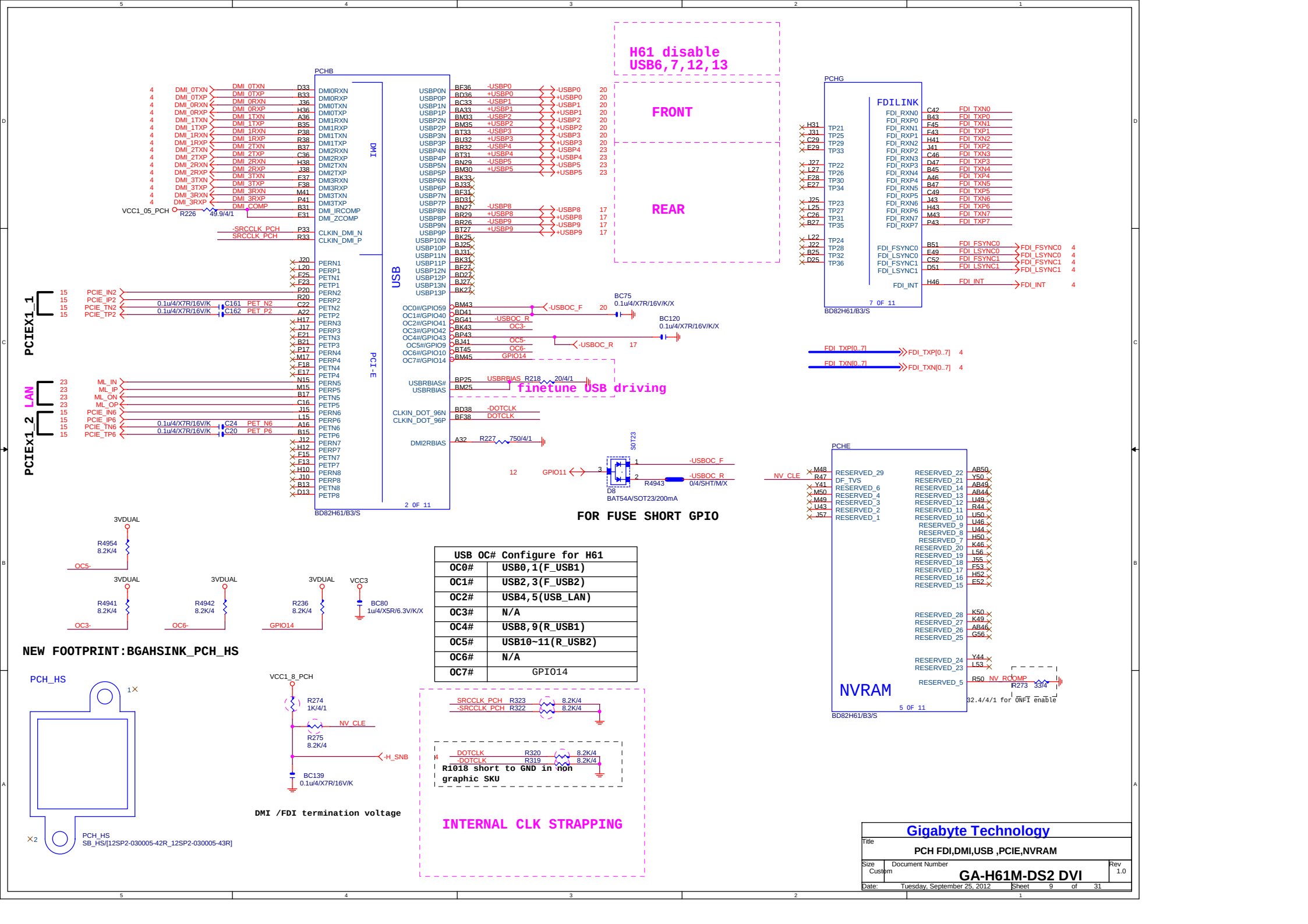
Gigabyte Technology	
Title	CPU LGA1156-C
Size: Custom	Document Number: GA-H61M-DS2 DVI
Rev	1.0
Date: Tuesday, September 25, 2012	Sheet 6 of 31


DDR3_1
DDRVTT
VTT
VSS
VDD
FREE
RSVD
ODT1
ODT0
NC/PAR_IN
NC/ERR_OUT
NC/TEST4
CB0
CB1
CB2
CB3
CB4
CB5
CB6
CB7
DQS0
DQS0*
DQS1
DQS1*
DQS2
DQS2*
DQS3
DQS3*
DQS4
DQS4*
DQS5
DQS5*
DQS6
DQS6*
DQS7
DQS7*
DQS8
DQS8*
DM0/DQS9
NC/DQS9*
DM1/DQS10
NC/DQS10*
DM2/DQS11
NC/DQS11*
DM3/DQS12
NC/DQS12*
DM4/DQS13
NC/DQS13*
DM5/DQS14
NC/DQS14*
DM6/DQS15
NC/DQS15*
DM7/DQS16
NC/DQS16*
DM8/DQS17
NC/DQS17*
DQ0
DQ1
DQ2
DQ3
DQ4
DQ5
DQ6
DQ7
DQ8
DQ9
DQ10
DQ11
DQ12
DQ13
DQ14
DQ15
DQ16
DQ17
DQ18
DQ19
DQ20
DQ21
DQ22
DQ23
DQ24
DQ25
DQ26
DQ27
DQ28
DQ29
DQ30
DQ31
DQ32
DQ33
DQ34
DQ35
DQ36
DQ37
DQ38
DQ39
DQ40
DQ41
DQ42
DQ43
DQ44
DQ45
DQ46
DQ47
DQ48
DQ49
DQ50
DQ51
DQ52
DQ53
DQ54
DQ55
DQ56
DQ57
DQ58
DQ59
DQ60
DQ61
DQ62
DQ63
MDQT_A[0..1]
-DQSA[0..7]
DQSA[0..7]
MDA[0..63]
DDR_15V
TC3
0.1u/4/X7R/16V/K
VCC3
VDDSPD
TC5 0.1u/4/X7R/16V/K VREF_DDRA
TC4 0.1u/4/X7R/16V/K VREF_DQDDRA
VREFCA
VREFDQ
SMBCLK
SMBDATA
SCL
SDA
SA1
SA0
SBAA2
SBAA1
SBAA0
BA2
BA1
BA0
CKEA1
CKEA0
CKE1
CKE0
-CSA1
-CSA0
S1*
S0*
-DCLKA1
DCLKA1
CK1/NU*
CK1/NU
-DCLKA0
DCLKA0
CK0*
CK0
MAAA[0..15]
A0
A1
A2
A3
A4
A5
A6
A7
A8
A9
A10/AP
A11
A12
A13
A14
A15
-DDR3_RST
-SCASA
-SRASA
-SWEA
RESET*
CAS*
RAS*
WE*
DDR3/240/BU/VA/D
R292
1K/4/1
VREF_DDRA
R293
1K/4/1
BC141
1u/4/X5R/6.3V/K/X
TR28
1K/4/1
VREF_DQDDRA
VREF_DQA
TR27
1K/4/1
BC140
1u/4/X5R/6.3V/K
-DDR3_RST
TC2
100p/4/NPO/50V/J
SMBDATA
SMBCLK
TC10
100p/4/NPO/50V/J/X
TC9
100p/4/NPO/50V/J/X
DDR15V Decouple
TEC7
560u/FP/D/6.3V/69/A/11m
TEC6
560u/FP/D/6.3V/69/A/11m
TBC21
1u/4/X5R/6.3V/K
TBC23
1u/4/X5R/6.3V/K//X
TBC29
1u/4/X5R/6.3V/K
TBC20
1u/4/X5R/6.3V/K/X
TBC16
1u/4/X5R/6.3V/K
TBC15
1u/4/X5R/6.3V/K/X
TBC14
1u/4/X5R/6.3V/K
TBC30
0.1u/4/Y5V/16V/Z/X
TBC24
0.1u/4/Y5V/16V/Z
TBC22
0.1u/4/Y5V/16V/Z/X
TBC27
0.1u/4/Y5V/16V/Z
TBC11
0.1u/4/Y5V/16V/Z/X
TBC12
0.1u/4/Y5V/16V/Z
TBC13
0.1u/4/Y5V/16V/Z/X
TBC32
0.1u/4/Y5V/16V/Z/X
TBC28
0.1u/4/Y5V/16V/Z/X
TBC31
0.1u/4/Y5V/16V/Z/X
DDRVTT Decouple
TBC35
22u/8/X5R/6.3V/M
TBC34
22u/8/X5R/6.3V/M/X
TBC19
1u/4/X5R/6.3V/K
TBC26
1u/4/X5R/6.3V/K/X
TBC33
0.1u/4/Y5V/16V/Z/X
TBC17
0.1u/4/Y5V/16V/Z/X
TBC18
0.1u/4/Y5V/16V/Z/X
Gigabyte Technology
Title
DDRIII CHANNEL A
Size
Custom
Document Number
GA-H61M-DS2 DVI
Rev
1.0
Date:
Sheet 7 of 31



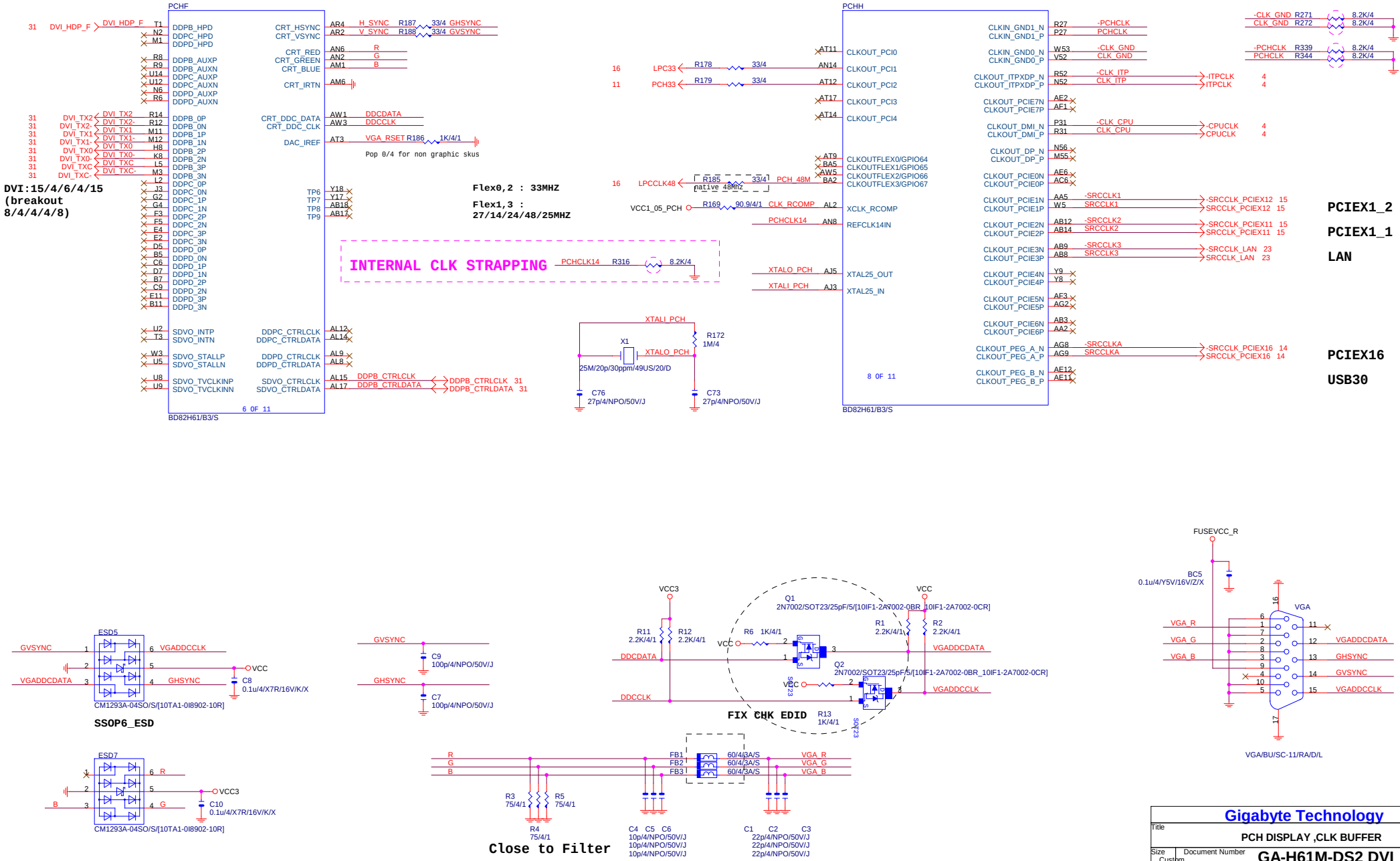
Gigabyte Confidential
Do not Copy

Gigabyte Technology		
Title: DDRIII CHANNEL B		
Size: Custom	Document Number: GA-H61M-DS2 DVI	Rev: 1.0
Date:	Sheet 8 of 31	

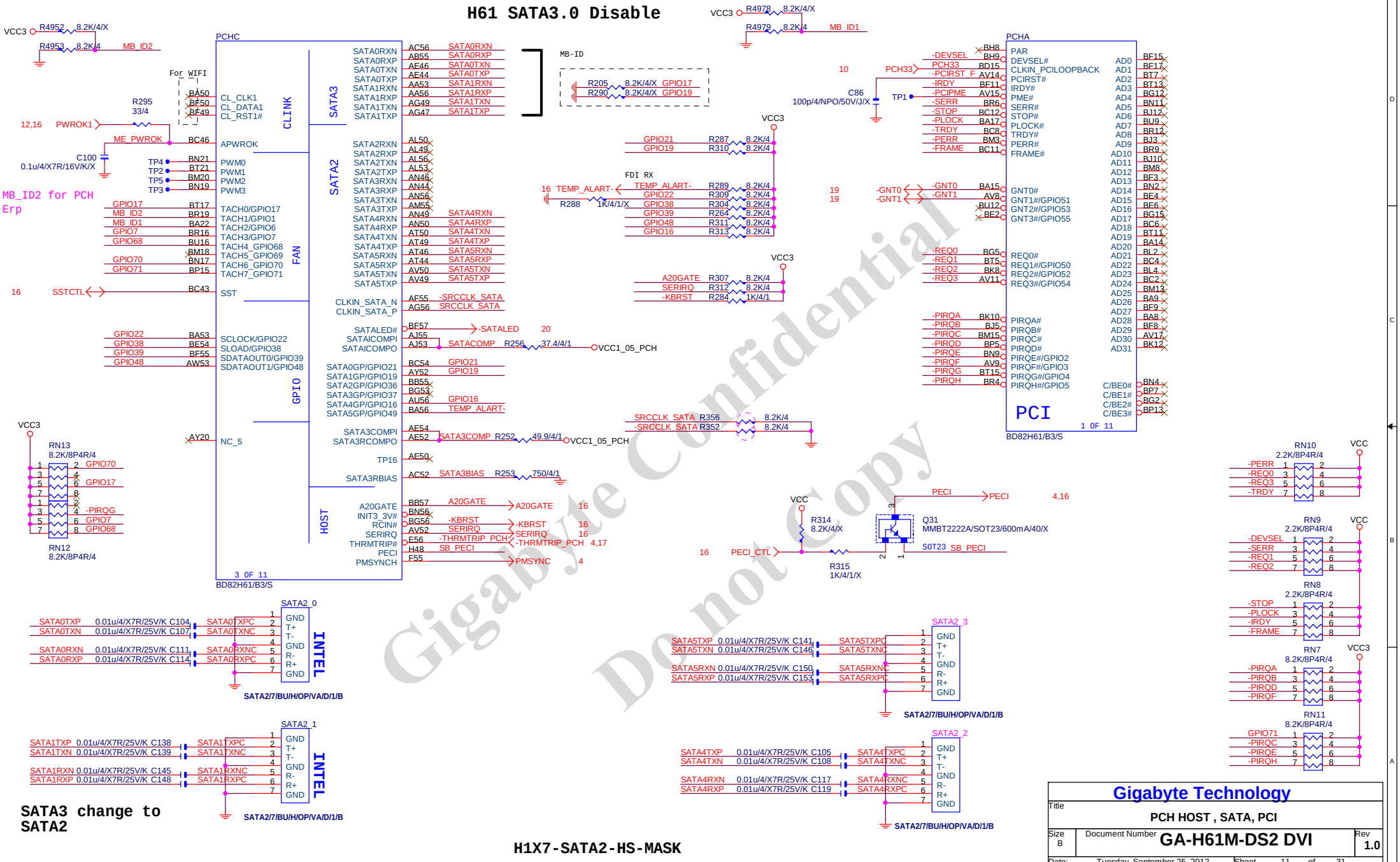


USB OC# Configure for H61	
OC0#	USB0,1(F_USB1)
OC1#	USB2,3(F_USB2)
OC2#	USB4,5(USB_LAN)
OC3#	N/A
OC4#	USB8,9(R_USB1)
OC5#	USB10~11(R_USB2)
OC6#	N/A
OC7#	GPIO14

Gigabyte Technology		
Title: PCH FDI,DMI,USB ,PCIE,NVRAM		
Size: Custom	Document Number: GA-H61M-DS2 DVI	Rev: 1.0
Date: Tuesday, September 25, 2012	Sheet 9 of 31	


PCHF
DDPB_HPD
DDPC_HPD
DDPD_HPD
DDPB_AUXP
DDPB_AUXN
DDPC_AUXP
DDPC_AUXN
DDPD_AUXP
DDPD_AUXN
DDPB_0P
DDPB_0N
DDPB_1P
DDPB_1N
DDPB_2P
DDPB_2N
DDPB_3P
DDPB_3N
DDPC_0P
DDPC_0N
DDPC_1P
DDPC_1N
DDPC_2P
DDPC_2N
DDPC_3P
DDPC_3N
DDPD_0P
DDPD_0N
DDPD_1P
DDPD_1N
DDPD_2P
DDPD_2N
DDPD_3P
DDPD_3N
SDVO_INTP
SDVO_INTN
SDVO_STALLP
SDVO_STALLN
SDVO_TVCLKINP
SDVO_TVCLKINN
CRT_HSYNC
CRT_VSYNC
CRT_RED
CRT_GREEN
CRT_BLUE
CRT_IRTN
CRT_DDC_DATA
CRT_DDC_CLK
DAC_IREF
DDPC_CTRLCLK
DDPC_CTRLDATA
DDPD_CTRLCLK
DDPD_CTRLDATA
SDVO_CTRLCLK
SDVO_CTRLDATA
6 OF 11
BD82H61/B3/S
DVI:15/4/6/4/15
(breakout
8/4/4/4/8)
Flex0,2 : 33MHZ
Flex1,3 :
27/14/24/48/25MHZ
INTERNAL CLK STRAPPING
PCHCLK14
XTALO_PCH
XTALI_PCH
X1
25M/20p/30ppm/49US/20/D
C76
27p/4/NPO/50V/J
C73
27p/4/NPO/50V/J
PCHH
CLKOUT_PCI0
CLKOUT_PCI1
CLKOUT_PCI2
CLKOUT_PCI3
CLKOUT_PCI4
CLKOUTFLEX0/GPIO64
CLKOUTFLEX1/GPIO65
CLKOUTFLEX2/GPIO66
CLKOUTFLEX3/GPIO67
XCLK_RCOMP
REFCLK14IN
XTAL25_OUT
XTAL25_IN
CLKIN_GND1_N
CLKIN_GND1_P
CLKIN_GND0_N
CLKIN_GND0_P
CLKOUT_ITPXDP_N
CLKOUT_ITPXDP_P
CLKOUT_PCIE7N
CLKOUT_PCIE7P
CLKOUT_DMI_N
CLKOUT_DMI_P
CLKOUT_DP_N
CLKOUT_DP_P
CLKOUT_PCIE0N
CLKOUT_PCIE0P
CLKOUT_PCIE1N
CLKOUT_PCIE1P
CLKOUT_PCIE2N
CLKOUT_PCIE2P
CLKOUT_PCIE3N
CLKOUT_PCIE3P
CLKOUT_PCIE4N
CLKOUT_PCIE4P
CLKOUT_PCIE5N
CLKOUT_PCIE5P
CLKOUT_PCIE6N
CLKOUT_PEG_A_N
CLKOUT_PEG_A_P
CLKOUT_PEG_B_N
CLKOUT_PEG_B_P
8 OF 11
BD82H61/B3/S
PCIEX1_2
PCIEX1_1
LAN
PCIEX16
USB30
ESD5
CM1293A-04SO/S/[10TA1-0I8902-10R]
SSOP6_ESD
ESD7
CM1293A-04SO/S/[10TA1-0I8902-10R]
Q1
2N7002/SOT23/25pF/5/[10IF1-2A7002-0BR_10IF1-2A7002-0CR]
Q2
2N7002/SOT23/25pF/5/[10IF1-2A7002-0BR_10IF1-2A7002-0CR]
FIX CHK EDID
Close to Filter
VGA
VGA/BU/SC-11/RA/D/L
Gigabyte Technology
PCH DISPLAY ,CLK BUFFER
GA-H61M-DS2 DVI
Rev 1.0
Tuesday, September 25, 2012
Sheet 10 of 31

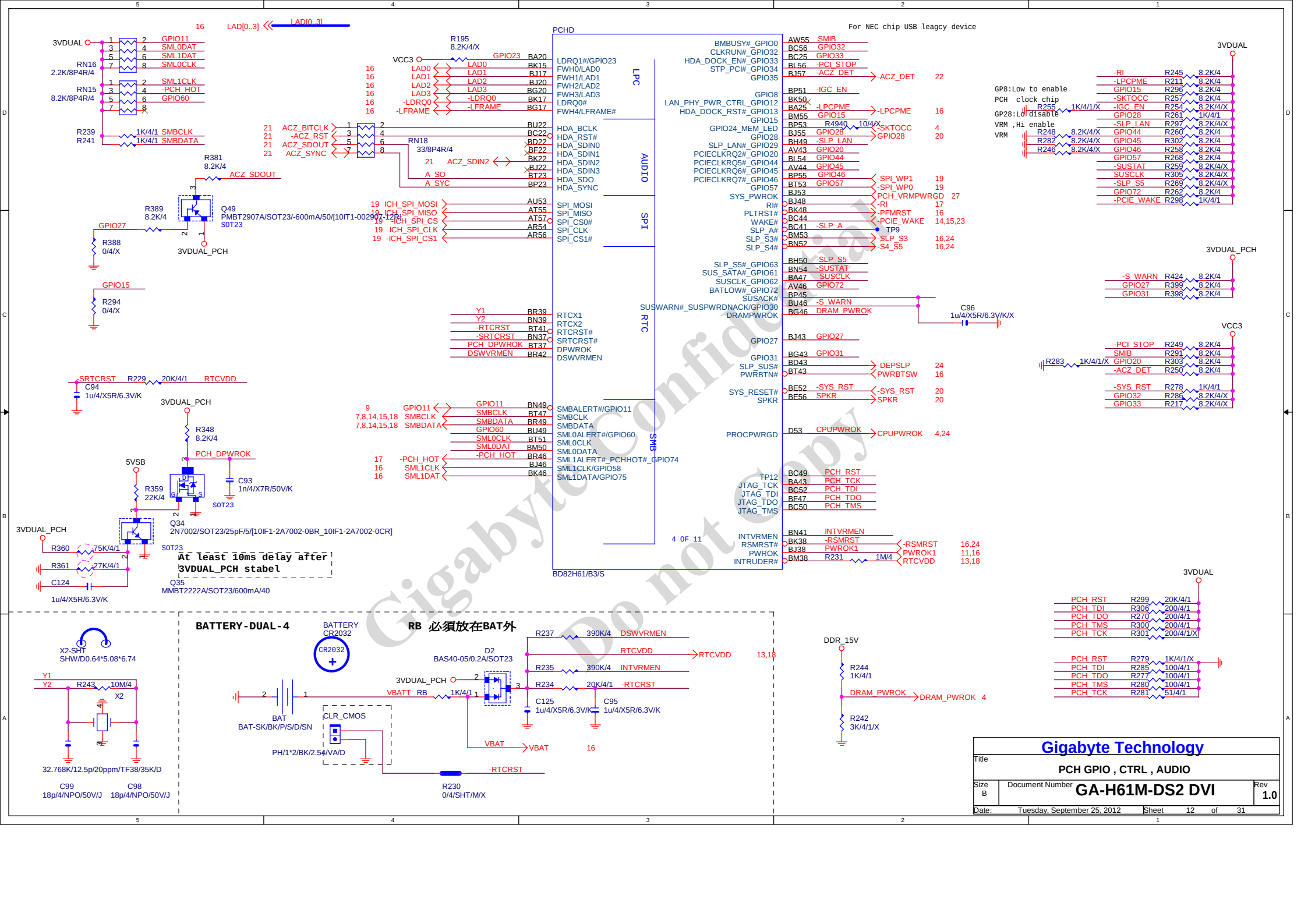
H61 SATA3.0 Disable

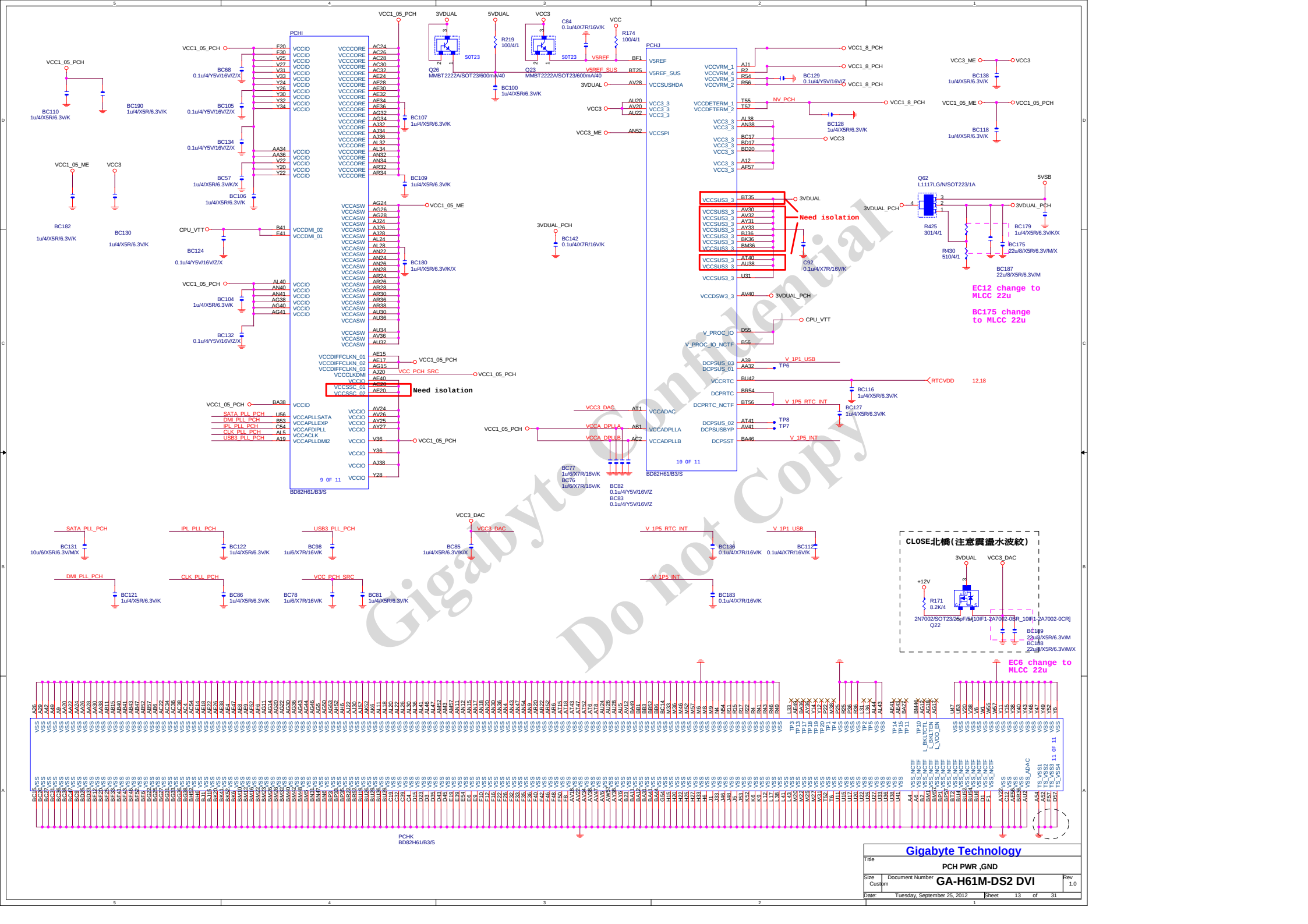


Gigabyte Technology

Title	PCH HOST , SATA, PCI	
Size B	Document Number GA-H61M-DS2 DVI	Rev 1.0
Date: Tuesday, September 25, 2012	Sheet 11 of 31	

H1X7-SATA2-HS-MASK


PCHD
LPC
AUDIO
SPI
RTC
SMB
BD82H61/B3/S
For NEC chip USB leagcy device
LAD[0..3]
3VDUAL
3VDUAL_PCH
VCC3
5VSB
DDR_15V
RN16 2.2K/8P4R/4
RN15 8.2K/8P4R/4
RN18 33/8P4R/4
R195 8.2K/4/X
R381 8.2K/4
R389 8.2K/4
R388 0/4/X
R294 0/4/X
R348 8.2K/4
R359 22K/4
R360 75K/4/1
R361 27K/4/1
C124 1u/4/X5R/6.3V/K
C94 1u/4/X5R/6.3V/K
C93 1n/4/X7R/50V/K
C96 1u/4/X5R/6.3V/K/X
Q49 PMBT2907A/SOT23/-600mA/50/[10IT1-002907-12RI] S0T23
Q34 2N7002/SOT23/25pF/5/[10IF1-2A7002-0BR_10IF1-2A7002-0CR] S0T23
Q35 MMBT2222A/SOT23/600mA/40
At least 10ms delay after 3VDUAL_PCH stabel
BATTERY-DUAL-4
BATTERY CR2032
RB 必須放在BAT外
D2 BAS40-05/0.2A/SOT23
X2-SHT SHW/D0.64*5.08*6.74
X2 32.768K/12.5p/20ppm/TF38/35K/D
C99 18p/4/NPO/50V/J
C98 18p/4/NPO/50V/J
BAT BAT-SK/BK/P/S/D/SN
CLR_CMOS PH/1*2/BK/2.54/VA/D
R230 0/4/SHT/M/X
R237 390K/4
R235 390K/4
R234 20K/4/1
C125 1u/4/X5R/6.3V/K
C95 1u/4/X5R/6.3V/K
R244 1K/4/1
R242 3K/4/1/X
R231 1M/4
GP8:Low to enable PCH clock chip
GP28:Lo disable VRM ,Hi enable
VRM
Gigabyte Technology
PCH GPIO , CTRL , AUDIO
GA-H61M-DS2 DVI
Rev 1.0
Size B
Date: Tuesday, September 25, 2012
Sheet 12 of 31
Gigabyte Confidential
Do not copy


PCH1
PCHJ
PCHK
BD82H61/B3/S
9 OF 11
10 OF 11
11 OF 11
Need isolation
Need isolation
EC12 change to MLCC 22u
BC175 change to MLCC 22u
EC6 change to MLCC 22u
CLOSE北橋(注意震盪水波紋)
Q26 MMBT2222A/SOT23/600mA/40
Q23 MMBT2222A/SOT23/600mA/40
Q62 L1117LG/N/SOT223/1A
Q22 2N7002/SOT23/26pF/5x10IF1-2A7002-0ER_10IF1-2A7002-0CR]
Gigabyte Technology
PCH PWR ,GND
GA-H61M-DS2 DVI
Rev 1.0
Tuesday, September 25, 2012
13 of 31
Gigabyte Confidential
Do not Copy

EC6D_8D10MM-RH

PCIESLOT-164DN-P

3GIO_*16

PCIEX16

KEY

PCI-E/16X-164P/BU/LOWR EJECTOR

+12V

BC9 0.1u/4/X7R/16V/K/X

VCC3

BC16 0.1u/4/X7R/16V/K

BC13 0.1u/4/X7R/16V/K/X

BC14 0.1u/4/X7R/16V/K

BC15 0.1u/4/X7R/16V/K/X

RN5 0/8P4R/0402/SHT/X

RN3 0/8P4R/4/X

EC18 270u/FP/D/16V/8C/A/10m

X16_+12V

3VDUAL

R24 0/4/SHT/M/X

R16 0/4/SHT/M/X

R19 0/4/SHT/M/X

C15 22p/4/NPO/50V/J/X

-PCIE_RST

-PCIE_WAKE

SMBCLK

SMBDATA

SRCCLK_PCIEX16 10

-SRCCLK_PCIEX16 10

EXP_A_RXP[0..15] 4

EXP_A_RXN[0..15] 4

EXP_A_TXP[0..15] 4

EXP_A_TXN[0..15] 4

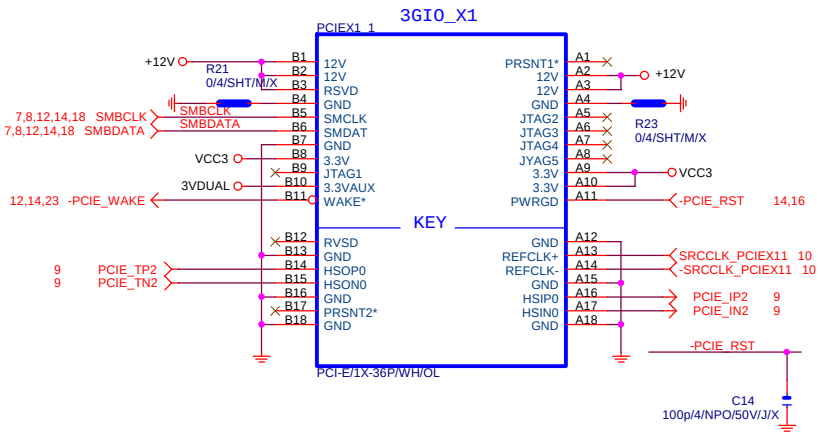
EXP_A_TXP0	C22	0.22u/4/X5R/6.3V/K EXP_A_TXP0C
EXP_A_TXN0	C26	0.22u/4/X5R/6.3V/K EXP_A_TXN0C
EXP_A_TXP1	C29	0.22u/4/X5R/6.3V/K EXP_A_TXP1C
EXP_A_TXN1	C31	0.22u/4/X5R/6.3V/K EXP_A_TXN1C
EXP_A_TXP2	C33	0.22u/4/X5R/6.3V/K EXP_A_TXP2C
EXP_A_TXN2	C35	0.22u/4/X5R/6.3V/K EXP_A_TXN2C
EXP_A_TXP3	C37	0.22u/4/X5R/6.3V/K EXP_A_TXP3C
EXP_A_TXN3	C40	0.22u/4/X5R/6.3V/K EXP_A_TXN3C
EXP_A_TXP4	C48	0.22u/4/X5R/6.3V/K EXP_A_TXP4C
EXP_A_TXN4	C49	0.22u/4/X5R/6.3V/K EXP_A_TXN4C
EXP_A_TXP5	C52	0.22u/4/X5R/6.3V/K EXP_A_TXP5C
EXP_A_TXN5	C54	0.22u/4/X5R/6.3V/K EXP_A_TXN5C
EXP_A_TXP6	C56	0.22u/4/X5R/6.3V/K EXP_A_TXP6C
EXP_A_TXN6	C58	0.22u/4/X5R/6.3V/K EXP_A_TXN6C
EXP_A_TXP7	C60	0.22u/4/X5R/6.3V/K EXP_A_TXP7C
EXP_A_TXN7	C61	0.22u/4/X5R/6.3V/K EXP_A_TXN7C
EXP_A_TXP8	C63	0.22u/4/X5R/6.3V/K EXP_A_TXP8C
EXP_A_TXN8	C64	0.22u/4/X5R/6.3V/K EXP_A_TXN8C
EXP_A_TXP9	C66	0.22u/4/X5R/6.3V/K EXP_A_TXP9C
EXP_A_TXN9	C67	0.22u/4/X5R/6.3V/K EXP_A_TXN9C
EXP_A_TXP10	C69	0.22u/4/X5R/6.3V/K EXP_A_TXP10C
EXP_A_TXN10	C70	0.22u/4/X5R/6.3V/K EXP_A_TXN10C
EXP_A_TXP11	C71	0.22u/4/X5R/6.3V/K EXP_A_TXP11C
EXP_A_TXN11	C72	0.22u/4/X5R/6.3V/K EXP_A_TXN11C
EXP_A_TXP12	C74	0.22u/4/X5R/6.3V/K EXP_A_TXP12C
EXP_A_TXN12	C75	0.22u/4/X5R/6.3V/K EXP_A_TXN12C
EXP_A_TXP13	C77	0.22u/4/X5R/6.3V/K EXP_A_TXP13C
EXP_A_TXN13	C78	0.22u/4/X5R/6.3V/K EXP_A_TXN13C
EXP_A_TXP14	C79	0.22u/4/X5R/6.3V/K EXP_A_TXP14C
EXP_A_TXN14	C80	0.22u/4/X5R/6.3V/K EXP_A_TXN14C
EXP_A_TXP15	C82	0.22u/4/X5R/6.3V/K EXP_A_TXP15C
EXP_A_TXN15	C83	0.22u/4/X5R/6.3V/K EXP_A_TXN15C

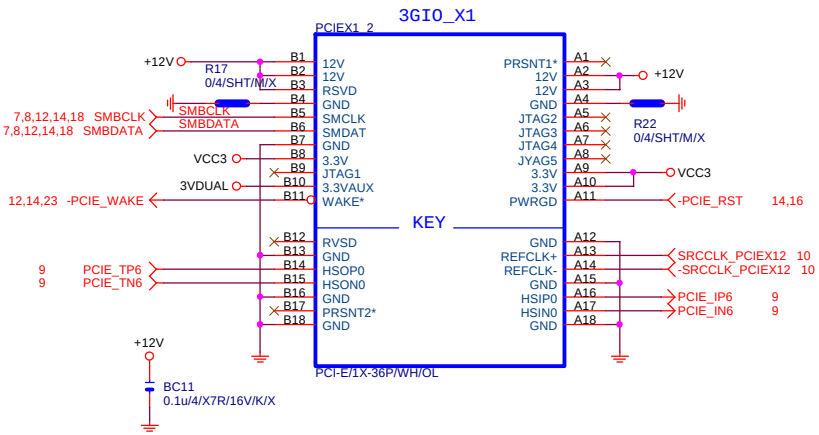
Gigabyte Confidential
Do not Copy

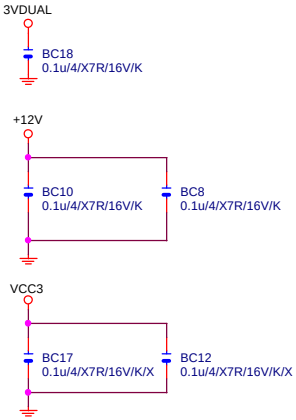
Title	PCI EXPRESS * 16
Size	Custom
Document Number	GA-H61M-DS2 DVI
Rev	1.0
Date:	Tuesday, September 25, 2012
Sheet	14 of 31

Gigabyte Technology

PCIE*1





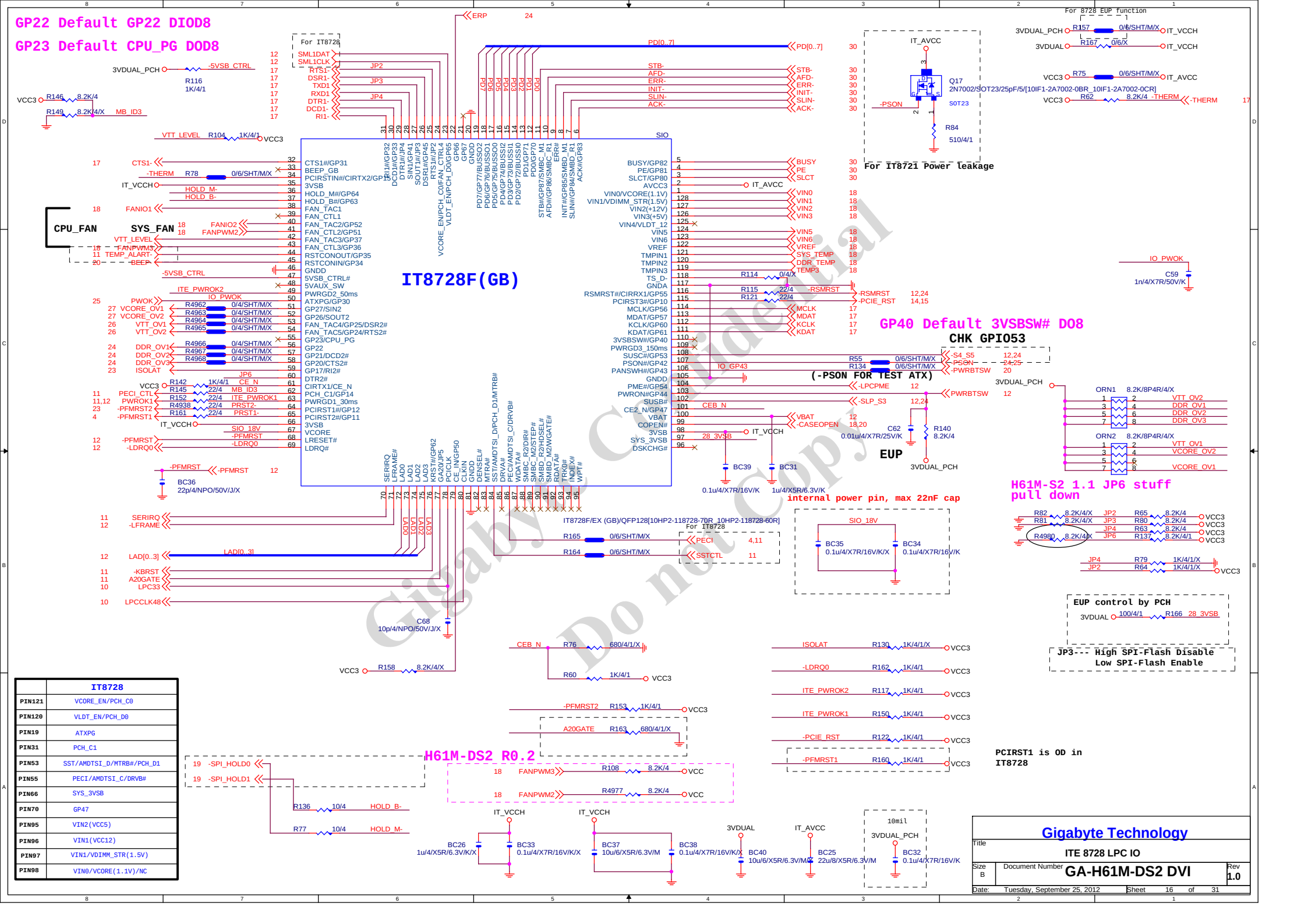


CLK GEN

Gigabyte Technology		
Title: PCIEX1,X2/CLK GEN		
Size: Custom	Document Number: GA-H61M-DS2 DVI	Rev: 1.0
Date: Tuesday, September 25, 2012	Sheet 15 of 31	

GP22 Default GP22 DIOD8

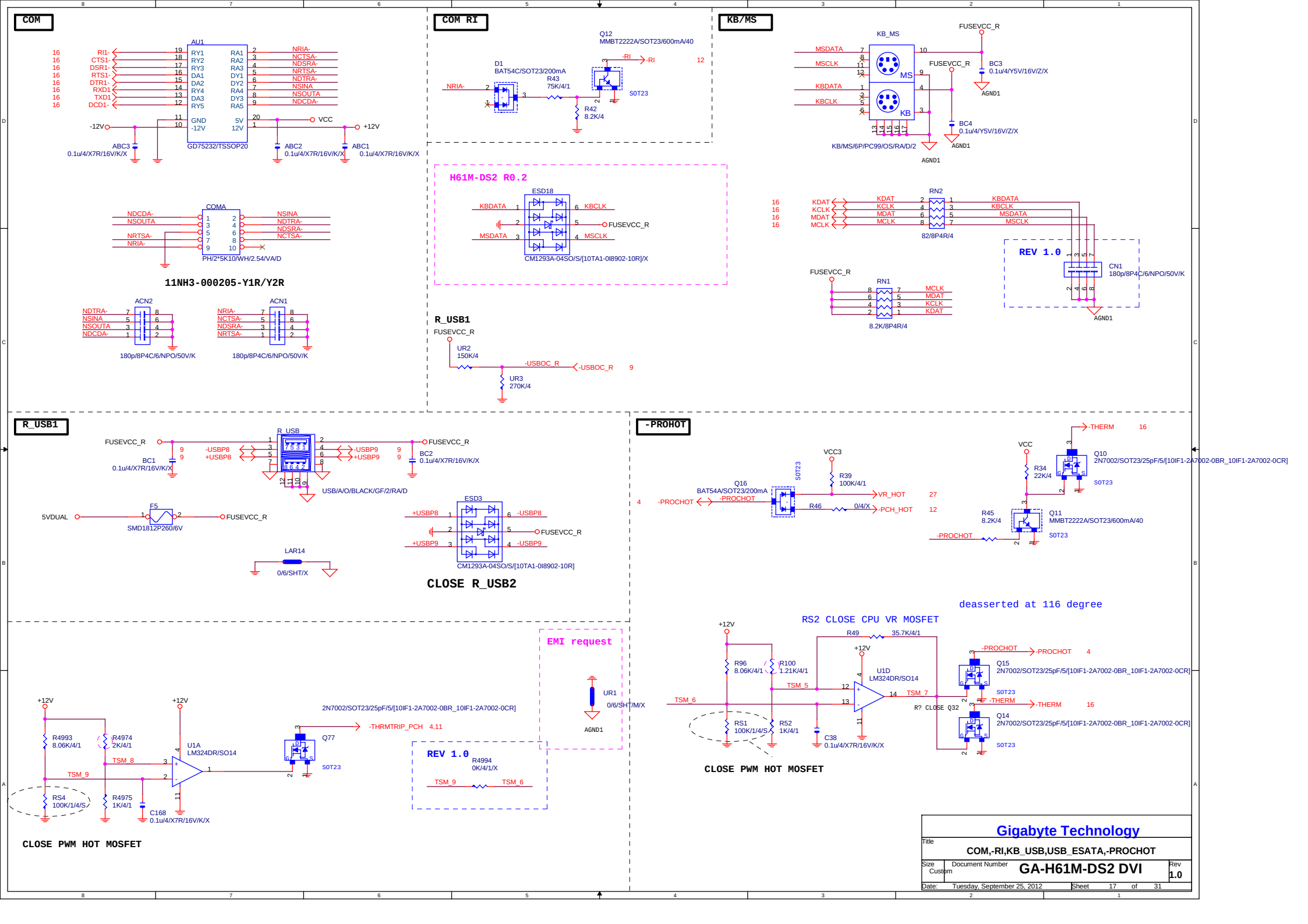
GP23 Default CPU_PG DOD8



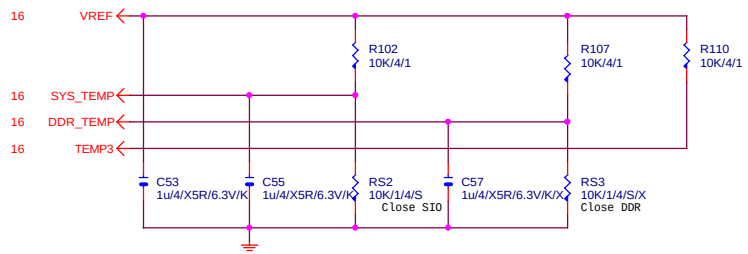
	IT8728
PIN121	VCORE_EN/PCH_C0
PIN120	VLDT_EN/PCH_D0
PIN19	ATXPG
PIN31	PCH_C1
PIN53	SST/AMDTSI_D/MTRB#/PCH_D1
PIN55	PECI/AMDTSI_C/DRVB#
PIN66	SYS_3VSB
PIN70	GP47
PIN95	VIN2(VCC5)
PIN96	VIN1(VCC12)
PIN97	VIN1/VDIMM_STR(1.5V)
PIN98	VIN0/VCORE(1.1V)/NC

Gigabyte Confidential Do not Copy

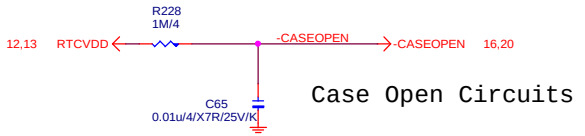
Gigabyte Technology			
Title: ITE 8728 LPC IO			
Size: B	Document Number: GA-H61M-DS2 DVI		Rev: 1.0
Date: Tuesday, September 25, 2012	Sheet 16 of 31		


COM
AU1
GD75232/TSSOP20
ABC3 0.1u/4/X7R/16V/K/X
ABC2 0.1u/4/X7R/16V/K/X
ABC1 0.1u/4/X7R/16V/K/X
COMA
PH/2*5K10/WH/2.54/VA/D
11NH3-000205-Y1R/Y2R
ACN2
ACN1
180p/8P4C/6/NPO/50V/K
COM RI
Q12
MMBT2222A/SOT23/600mA/40
D1
BAT54C/SOT23/200mA
R43
75K/4/1
R42
8.2K/4
SOT23
H61M-DS2 R0.2
ESD18
CM1293A-04SO/S/[10TA1-0I8902-10R]/X
R_USB1
UR2
150K/4
UR3
270K/4
KB/MS
KB_MS
MS
KB
KB/MS/6P/PC99/OS/RA/D/2
BC3
0.1u/4/Y5V/16V/Z/X
BC4
0.1u/4/Y5V/16V/Z/X
AGND1
RN2
82/8P4R/4
RN1
8.2K/8P4R/4
REV 1.0
CN1
180p/8P4C/6/NPO/50V/K
R_USB1
R_USB
USB/A/O/BLACK/GF/2/RA/D
BC1
0.1u/4/X7R/16V/K/X
BC2
0.1u/4/X7R/16V/K/X
F5
SMD1812P260/6V
LAR14
0/6/SHT/X
ESD3
CLOSE R_USB2
-PROHOT
Q16
BAT54A/SOT23/200mA
R39
100K/4/1
R46
0/4/X
Q10
2N7002/SOT23/25pF/5/[10IF1-2A7002-0BR_10IF1-2A7002-0CR]
R34
22K/4
R45
8.2K/4
Q11
MMBT2222A/SOT23/600mA/40
deasserted at 116 degree
RS2 CLOSE CPU VR MOSFET
R49
35.7K/4/1
R96
8.06K/4/1
R100
1.21K/4/1
U1D
LM324DR/SO14
RS1
100K/1/4/S
R52
1K/4/1
C38
0.1u/4/X7R/16V/K/X
Q15
Q14
R? CLOSE Q32
CLOSE PWM HOT MOSFET
EMI request
UR1
0/6/SHT/M/X
R4993
8.06K/4/1
R4974
2K/4/1
U1A
LM324DR/SO14
RS4
100K/1/4/S
R4975
1K/4/1
C168
0.1u/4/X7R/16V/K/X
Q77
-THRMTRIP_PCH 4,11
R4994
0K/4/1/X
CLOSE PWM HOT MOSFET
Gigabyte Technology
Title
COM,-RI,KB_USB,USB_ESATA,-PROCHOT
Size Custom
Document Number
GA-H61M-DS2 DVI
Rev 1.0
Date: Tuesday, September 25, 2012
Sheet 17 of 31

TEMP H/W MONITOR

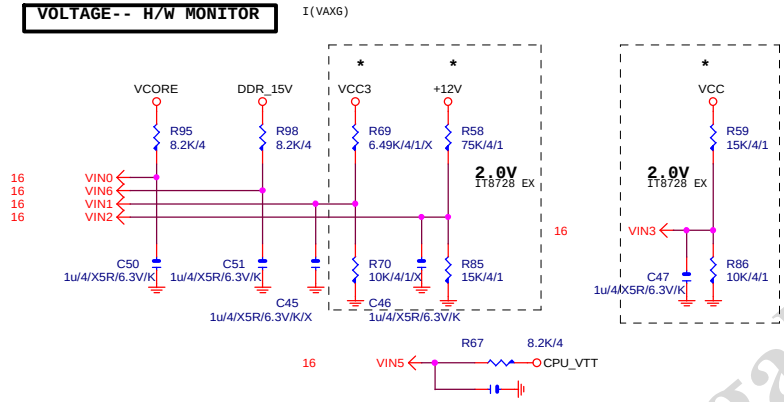


CASE OPEN

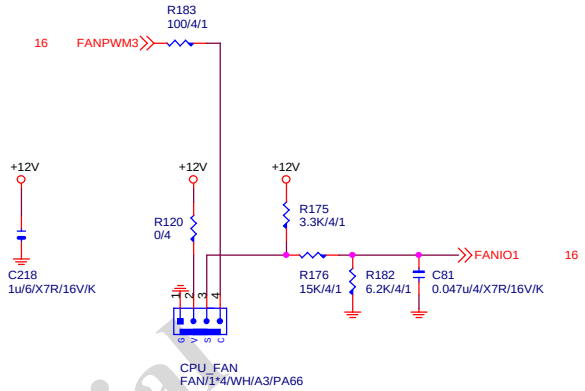


VOLTAGE-- H/W MONITOR

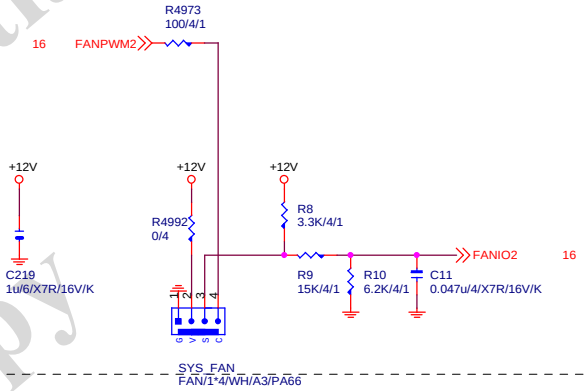
I(VAXG)

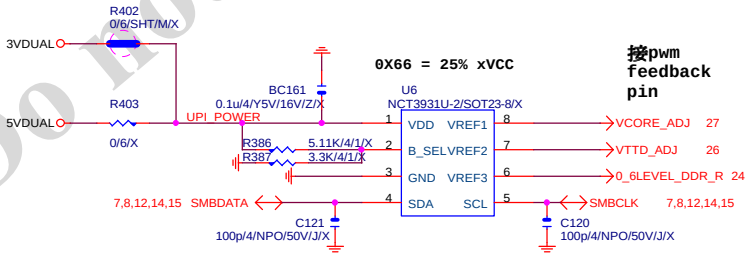


CPU SMART FAN



SYS SMART FAN

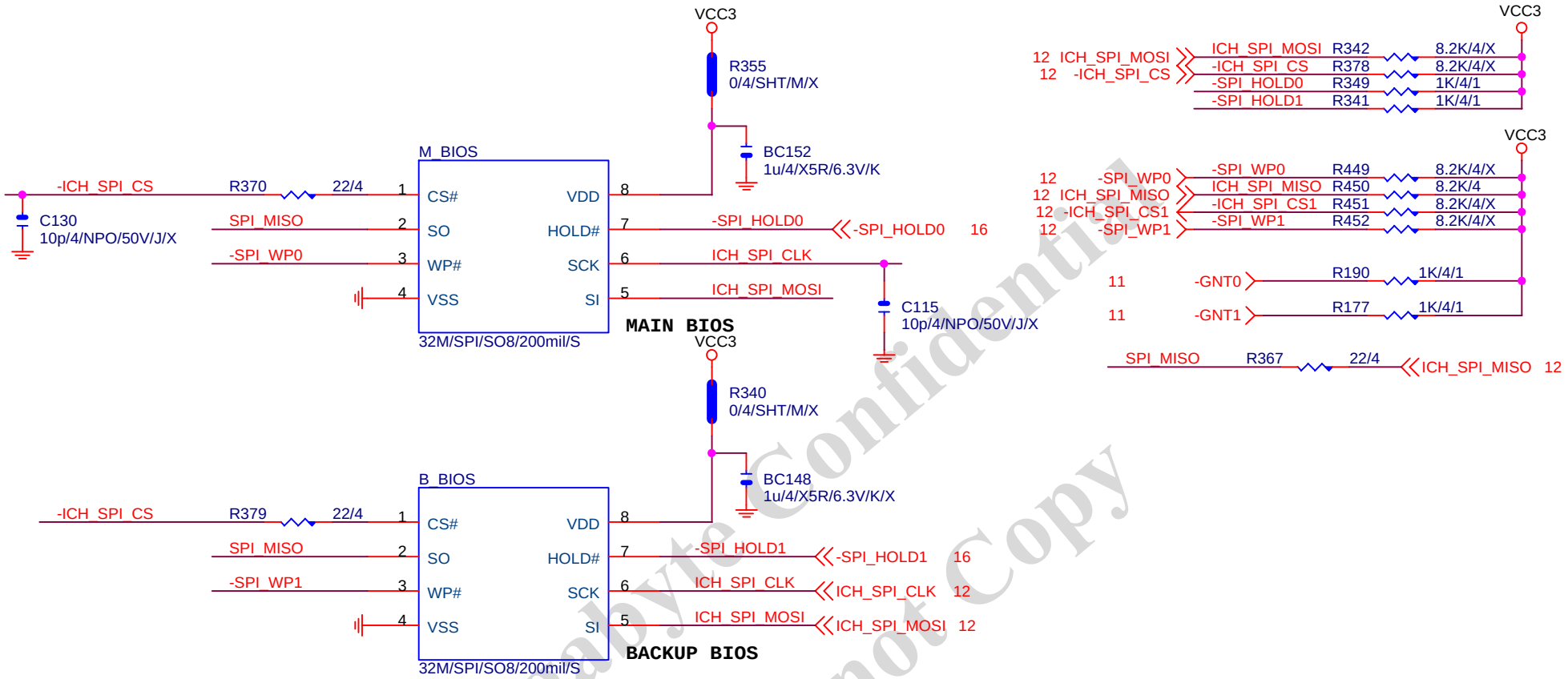




Gigabyte Confidential
Do Not Copy

Gigabyte Technology		
Title: HWM,FAN CTRL,OV		
Size: Custom	Document Number: GA-H61M-DS2 DVI	Rev: 1.0
Date: Tuesday, September 25, 2012	Sheet 18 of 31	

DUAL BIOS



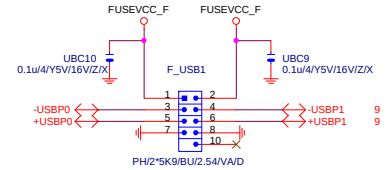
BOOT DEVICE	GNT1	GNT0
LPC	0	0
PCI	0	1
SPI	1	1

1 means floating
0 means PD 1K

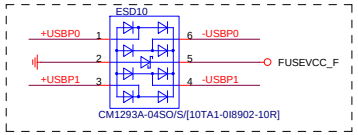
Gigabyte Confidential
Do not Copy

Gigabyte Technology		
Title: DUAL BIOS		
Size: A	Document Number: GA-H61M-DS2 DVI	Rev: 1.0
Date: Tuesday, September 25, 2012	Sheet 19 of 31	

FRONT USB1

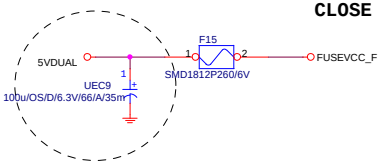


change to BLUE COLOR



Close to connector

CLOSE F_USB1

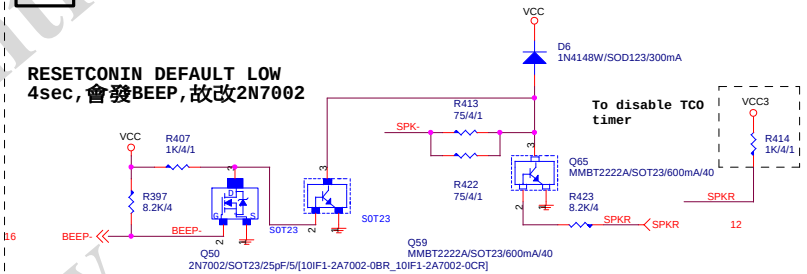


SATA LED

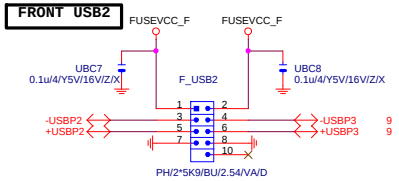


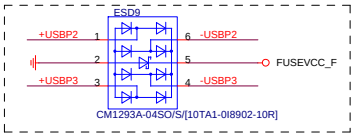
SPKR

RESETCONIN DEFAULT LOW
4sec,會發BEEP,故改2N7002



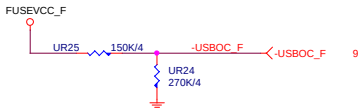
FRONT USB2



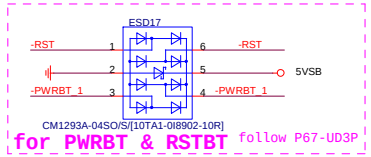


Close to connector

FRONT USB3

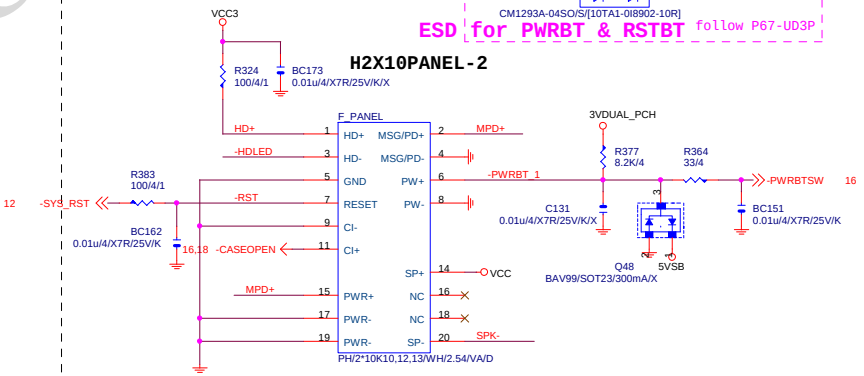


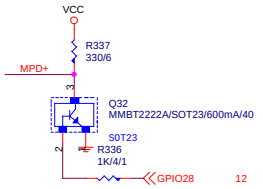
INTEL FRONT PANEL



ESD for PWRBT & RSTBT follow P67-UD3P

H2X10PANEL-2





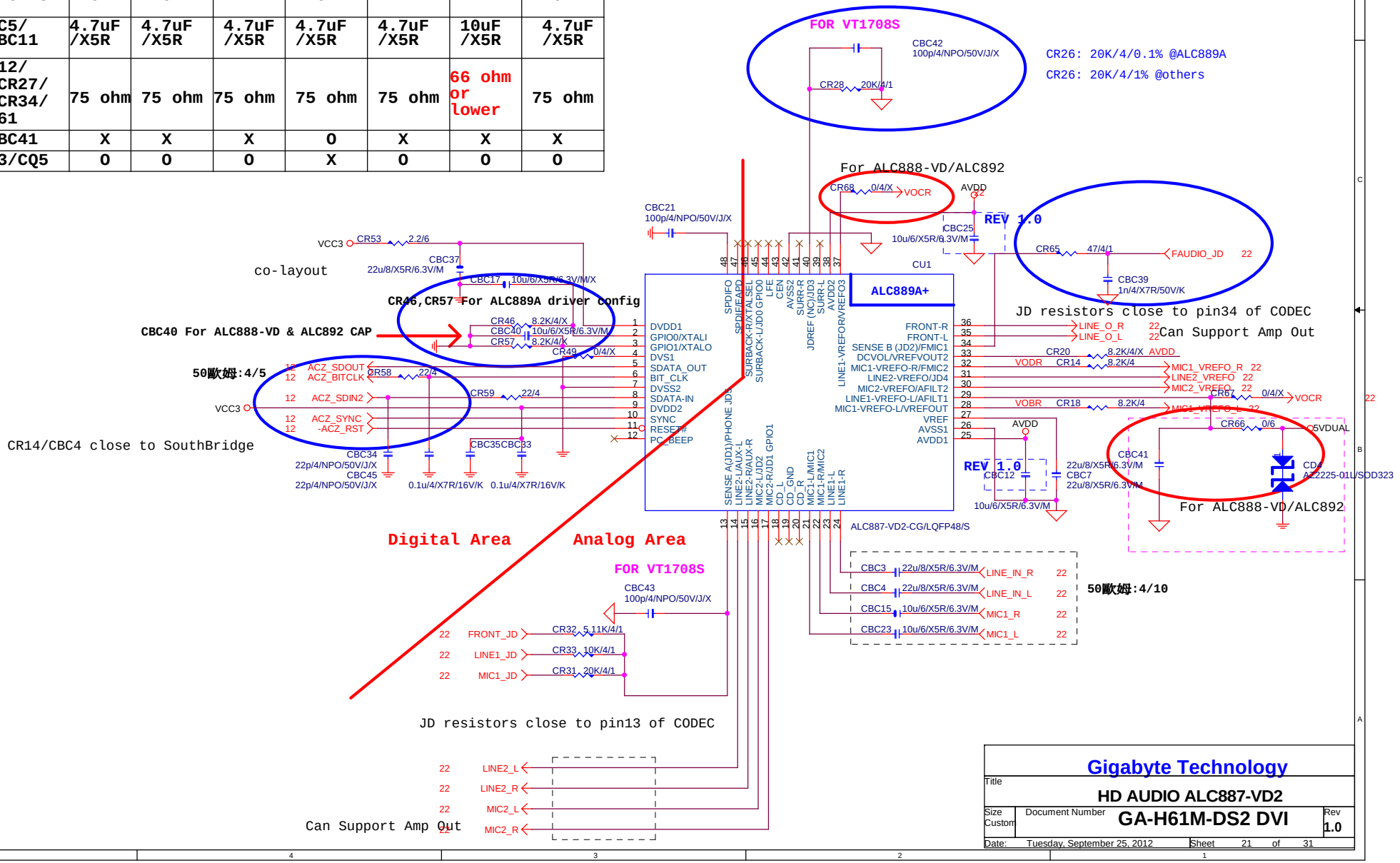
Gigabyte Confidential
Do not Copy

Gigabyte Technology		
Title: FP,F_USB,USB PWR,SPKR,SATA LED		
Size: Custom	Document Number: GA-H61M-DS2 DVI	Rev: 1.0
Date: Tuesday, September 25, 2012	Sheet 20 of 31	

AZALIA CODEC

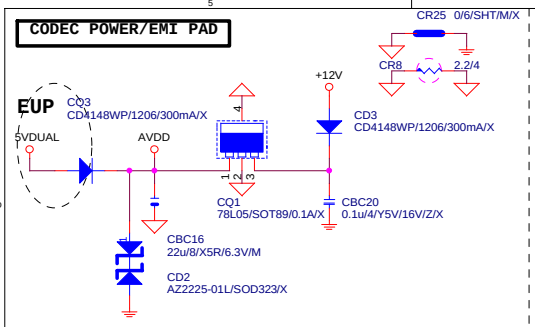
ALC883/ALC888-A/ALC888B/ALC888-VD/ALC892R/ALC889/ALC889A Colay

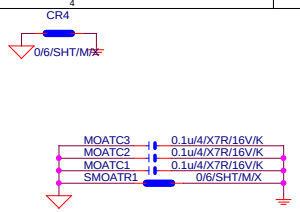
	ALC883	ALC888-VA	ALC888B	ALC888-VD	ALC892R	ALC889	ALC889A
CR46	X	X	X	X	X	X	O
CR57	X	X	X	X	X	X	O
CR49	O	O	X	X	X	O	O
CBC40	X	X	X	10uF/X5R	10uF/X5R	X	X
CR20	O	X	X	X	X	X	X
CR26	20K/1%	20K/1%	20K/1%	20K/1%	20K/1%	20K/1%	20K/0.1%
CR47	X	X	O	X	O	O	X
CR48	O	O	X	O	X	X	O
CBC2/CBC4/CBC5/ CBC6/CBC10/CBC11	4.7uF /X5R	4.7uF /X5R	4.7uF /X5R	4.7uF /X5R	4.7uF /X5R	10uF /X5R	4.7uF /X5R
CR1/CR3/CR10/CR12/ CR15/CR19/CR56/CR27/ CR55/CR37/CR28/CR34/ CR6/CR9/CR51/CR61	75 ohm	75 ohm	75 ohm	75 ohm	75 ohm	66 ohm or lower	75 ohm
CR66/CR68/CD3/CBC41	X	X	X	O	X	X	X
CR67/CD1/CD2/CQ3/CQ5	O	O	O	X	O	O	O



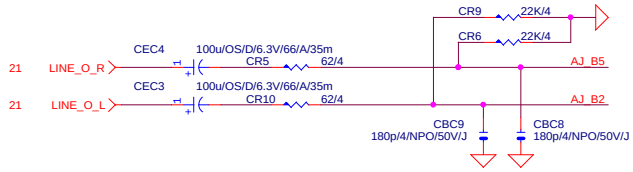
Gigabyte Technology		
Title	HD AUDIO ALC887-VD2	
Size: Custom	Document Number: GA-H61M-DS2 DVI	Rev: 1.0
Date: Tuesday, September 25, 2012	Sheet 21 of 31	

CODEC POWER/EMI PAD



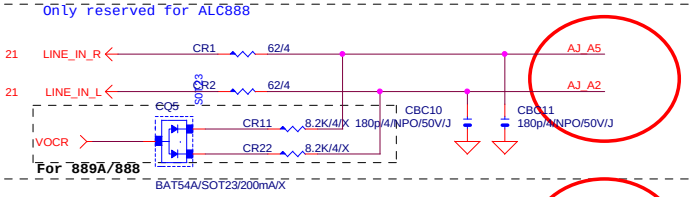


LINE-OUT



Only reserved for ALC888

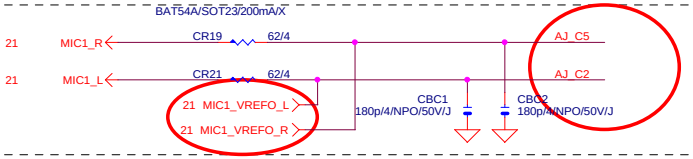
LINE-IN



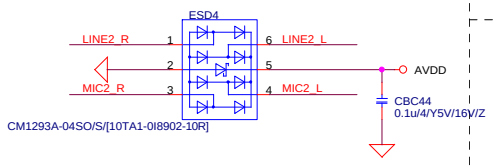
Verify MIC function in LINE-in

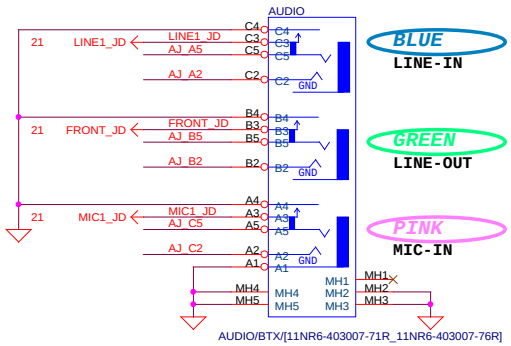
For 889A/888

MIC-IN

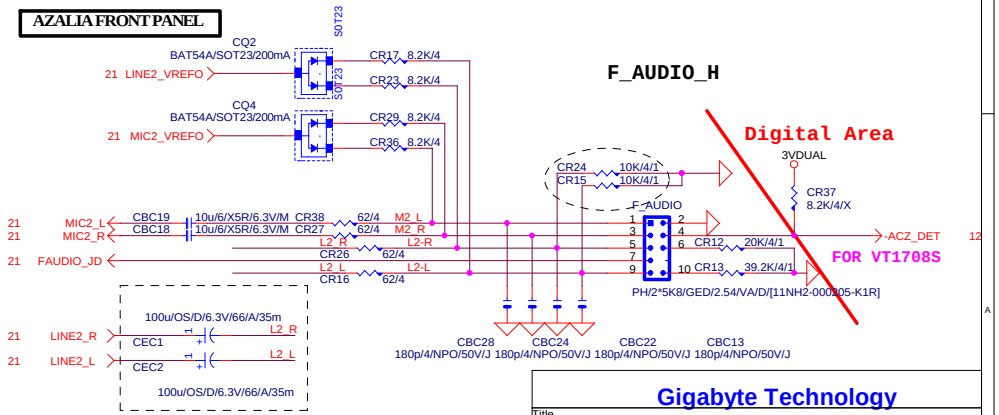


AZALIA JACK





AZALIA FRONT PANEL

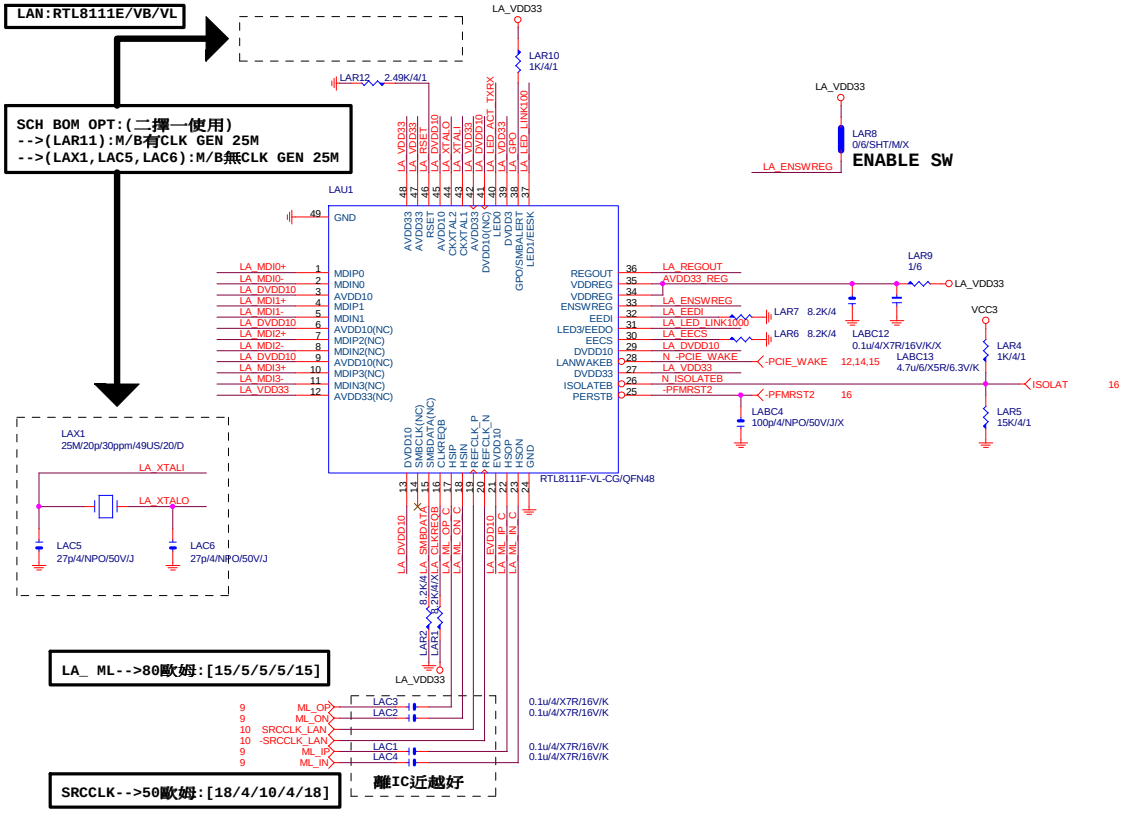


Gigabyte Technology

Title	AUDIO JACK	
Size: Custom	Document Number: GA-H61M-DS2 DVI	Rev: 1.0
Date: Tuesday, September 25, 2012	Sheet 22 of 31	

LAN:RTL8111E/VB/VL

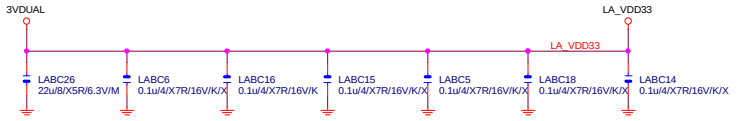
SCH BOM OPT:(二擇一使用)
-->(LAR11):M/B有CLK GEN 25M
-->(LAX1,LAC5,LAC6):M/B無CLK GEN 25M



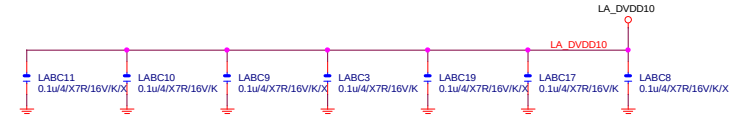
LA_ ML-->80歐姆:[15/5/5/5/15]

SRCCLK-->50歐姆:[18/4/10/4/18]

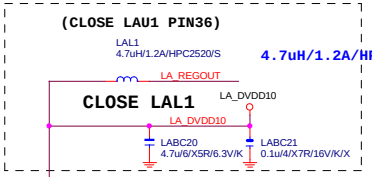
LAN POWER

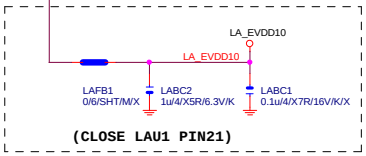


(CLOSE LAU1 PIN:12,27,39,42,47,48)



(CLOSE LAU1 PIN3,6,9,13,29,41,45)

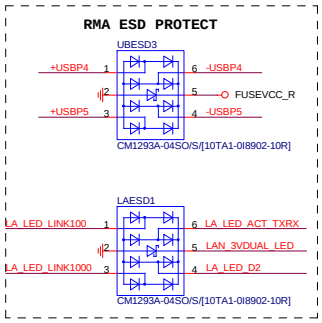




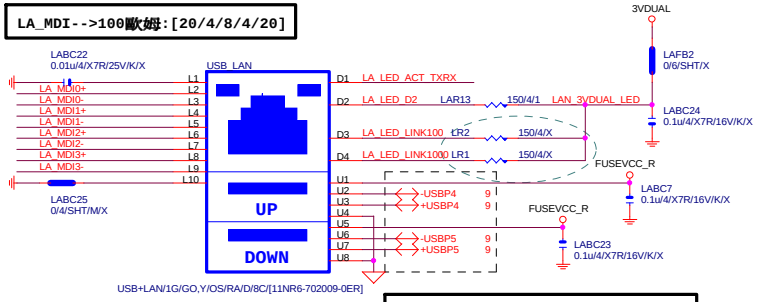
Power domain chart

	RTL8111E
AVDD33	3.3V
DVDD33	3.3V
VDDREG	3.3V
DVDD10	1.05V

USB_LAN CONNECTOR

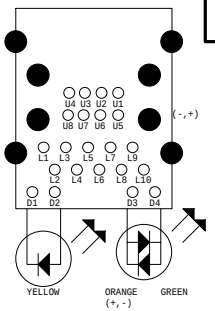


LA_MDI-->100歐姆:[20/4/8/4/20]

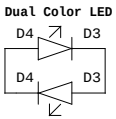


注意:USB PORT(目前:暫代6,7PORT)
USB-->90歐姆:[15/4.5/7.5/4.5/15]

注意:LAN LED PROTECT:(CO-LAYOUT)
1.ESD(6PIN):AOZ8902CIL/SOT23-6(DEFAULT)
2.SURGE(5PIN):AZ2025-04S/SOT23-5L

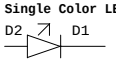


Dual Color LED



Green
Orange

Single Color LED



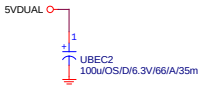
Yellow

料號	規格	廠商
11NR6-702009-0ER	1G LAN (12core)	UDE
11NR6-702009-91R	1G LAN(8 core)	FOXCONN
11NR6-702009-92R	1G LAN(8 core)	UDE
11NR6-702009-11R	1G LAN(12core/RED)	UDE
11NR6-702009-12R	1G LAN(8 core/RED)	FOXCONN

USB_LAN BOM區分:

1.(紅色/12CORE/三倍):USB+LAN/1G/GO,Y/OS/RA/D/1/RED
2.(黑色/12CORE):USB+LAN/1G/GO,Y/OS/RA/D/1
3.(黑色/8CORE):USB+LAN/1G/GO,Y/OS/RA/D/8C

USB X3 POWER



Close to connector

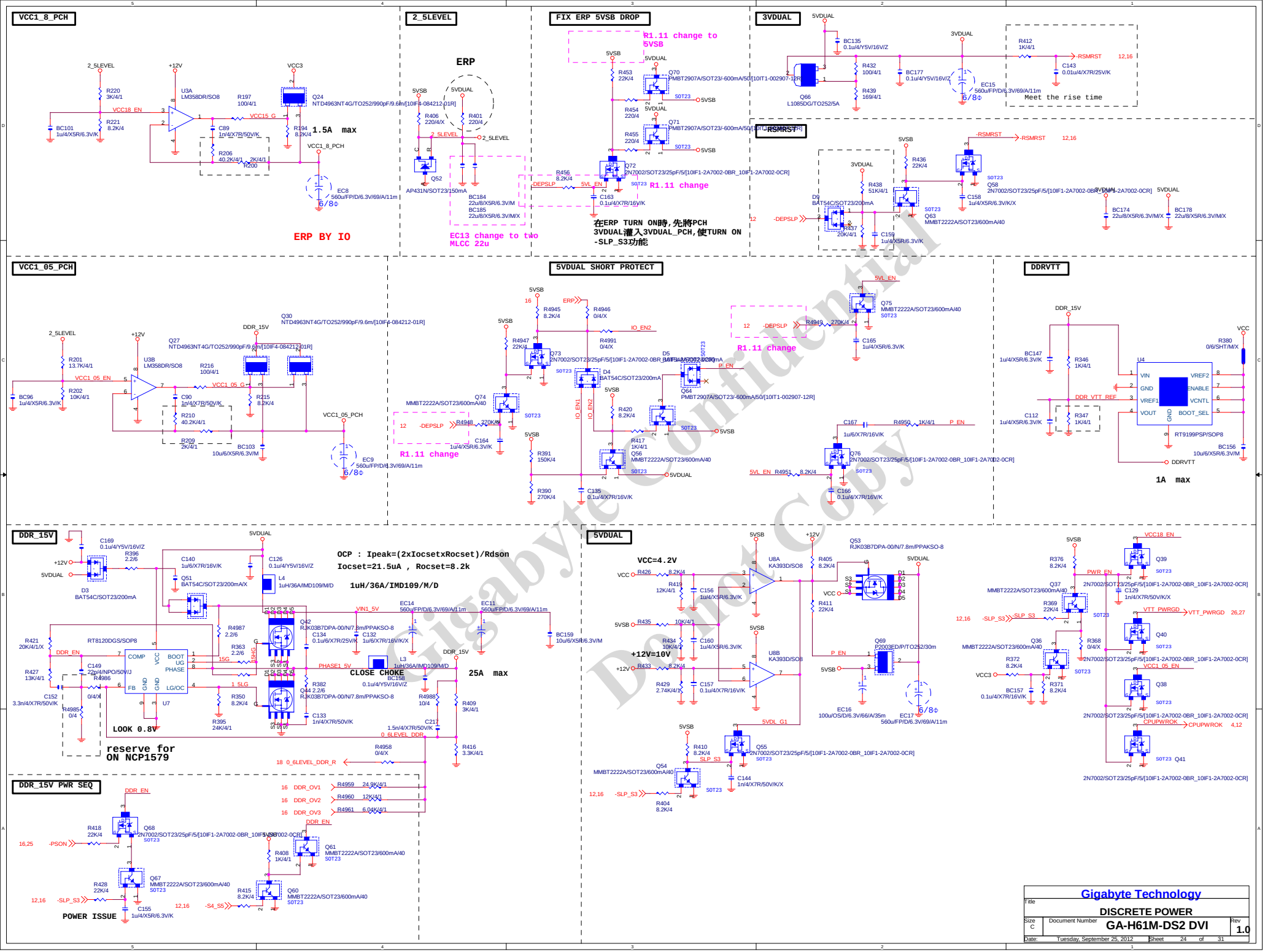
USB OC/FUSE PROTECT

Gigabyte Technology

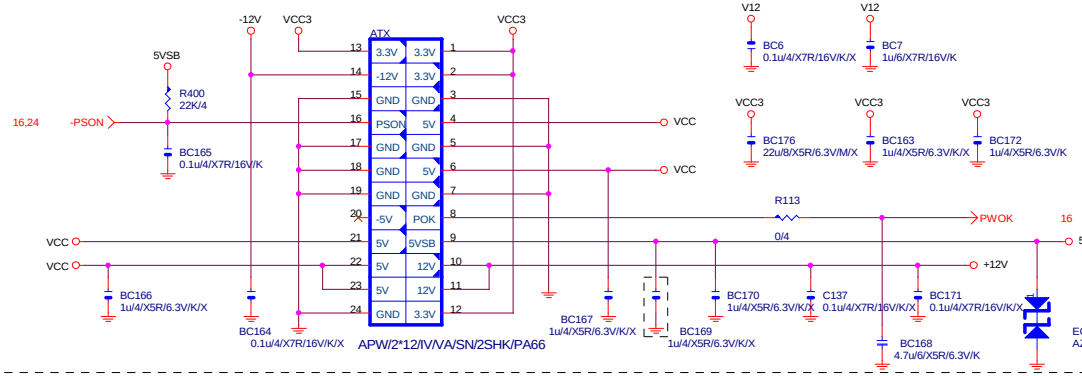
Title: LAN RTL8111E

Size: Custom | Document Number: GA-H61M-DS2 DVI | Rev: 1.0

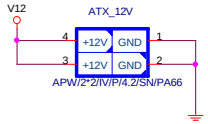
Date: Tuesday, September 25, 2012 | Sheet 23 of 31


VCC1_8_PCH
2_5LEVEL
FIX ERP 5VSB DROP
3VDUAL
ERP
R1.11 change to 5VSB
1.5A max
ERP BY IO
EC13 change to two MLCC 22u
R1.11 change
在ERP TURN ON時，先將PCH 3VDUAL灌入3VDUAL_PCH，使TURN ON -SLP_S3功能
Meet the rise time
RSMRST
VCC1_05_PCH
5VDUAL SHORT PROTECT
DDRVTT
1A max
DDR_15V
OCP : Ipeak=(2xIocsetxRocset)/Rdson
Iocset=21.5uA , Rocset=8.2k
1uH/36A/IMD109/M/D
CLOSE CHOKE
25A max
LOOK 0.8V
reserve for ON NCP1579
5VDUAL
VCC=4.2V
+12V=10V
DDR_15V PWR SEQ
POWER ISSUE
Gigabyte Confidential
Do not Copy
Gigabyte Technology
DISCRETE POWER
GA-H61M-DS2 DVI
Rev 1.0
Tuesday, September 25, 2012
Sheet 24 of 31

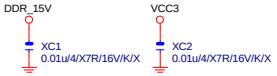
ATXX24 POWER CONNECTOR

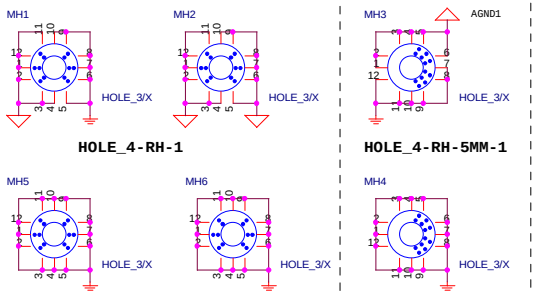


ATXX4 POWER CONNECTOR



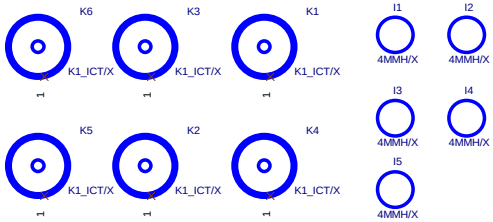
ATX_4-4





HOLE_4-RH-1

HOLE_4-RH-5MM-1

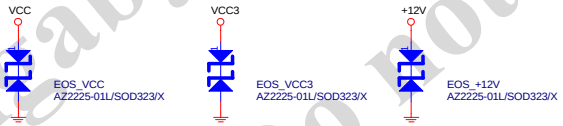


To prevent the 5VSB under loading when boot

5VDUAL1(USB PORT/DDRIII POWER)
5VDUAL(3VDUAL/OTHER)

-S_WARN-->5VDUAL1-->-S_ACK(PCH)-->-DEPSLP/-RSMRST-->5VDUAL-->3VDUAL

HOLE_4-RH-5MM-1



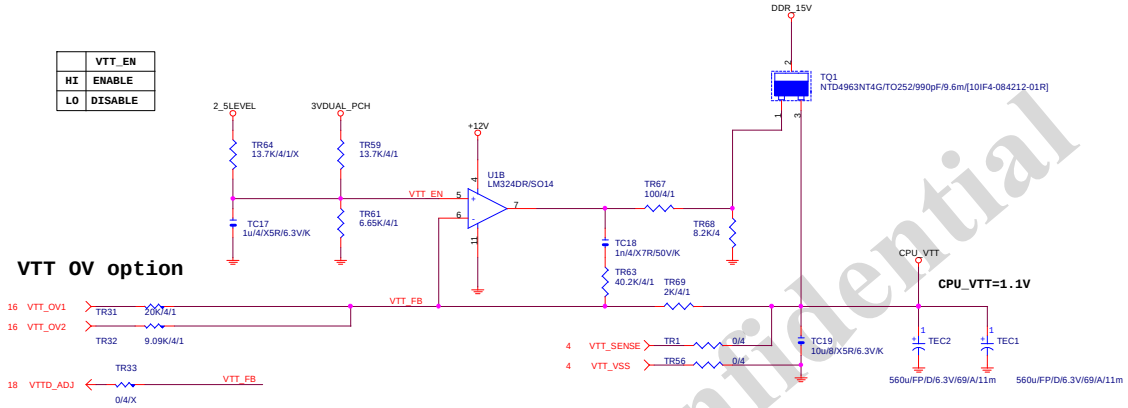
Gigabyte Confidential
Do not Copy

Gigabyte Technology		
Title: ATX CONNECTOR		
Size: Custom	Document Number: GA-H61M-DS2 DVI	Rev: 1.0
Date: Tuesday, September 25, 2012	Sheet 25 of 31	

CPU_VTT

	VTT_EN
HI	ENABLE
LO	DISABLE

VTT OV option

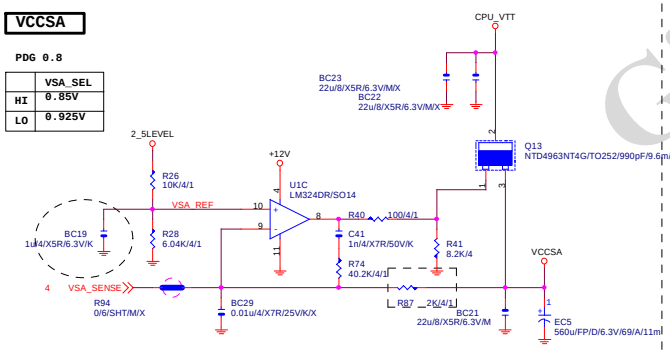


Gigabyte Confidential
Do not Copy

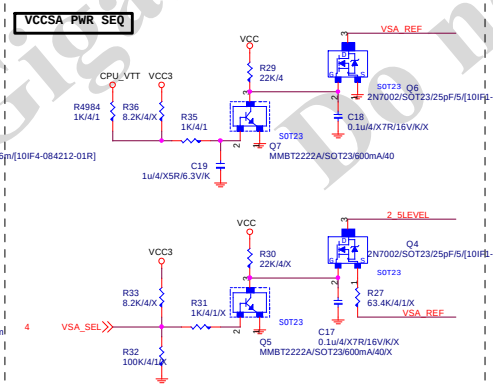
VCCSA

PDG 0.8

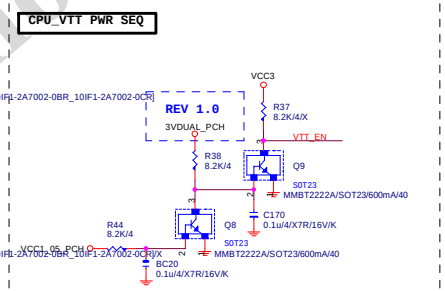
	VSA_SEL
HI	0.85V
LO	0.925V



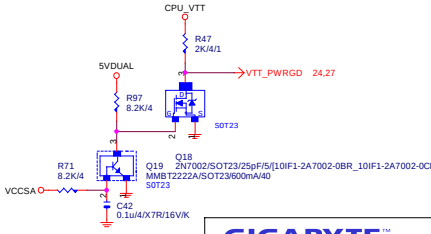
VCCSA PWR SEQ



CPU_VTT PWR SEQ

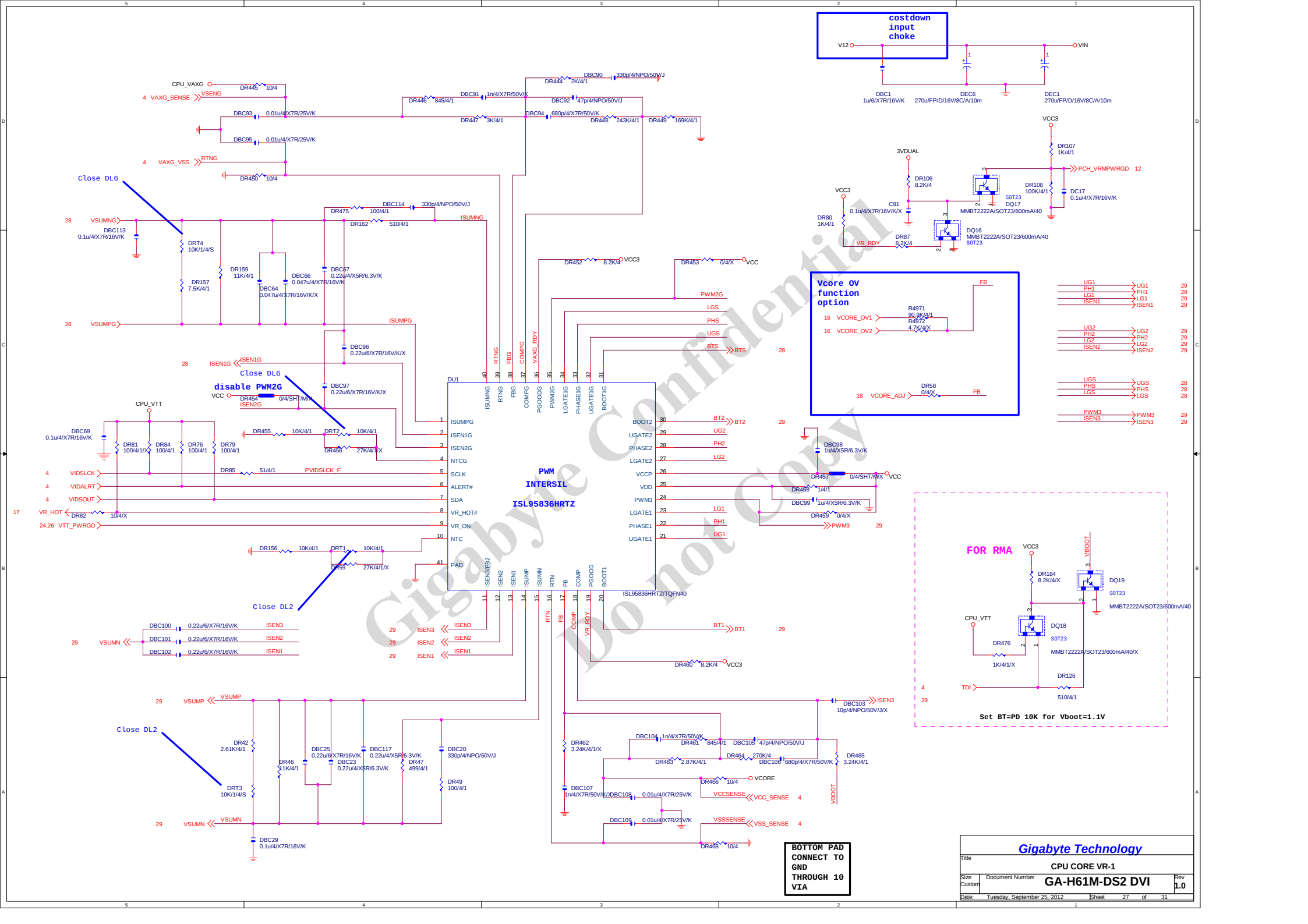


VTT_PWRGD



GIGABYTE™

Title	CPU_VTT PWM_ISL95870CRZ	
Size: Custom	Document Number: GA-H61M-DS2 DVI	Rev: 1.0
Date: Tuesday, September 25, 2012	Sheet 26 of 31	



Gigabyte Confidential
Do not Copy

VAXG

UGS
PHS
LGS

UGS 27
PHS 27
LGS 27

DR30
8.2K/4

DR28
2.2/6

UGS
UGS_1

PHS

LGS
DR35
0/6/SHT/M/X
LGS_1

DR31
2.2/6

DBC110
0.22u/6/X7R/16V/K

DC9
1n/4/X7R/50V/K

DR469
0/6/SHT/M/X

27 BTS

VIN

D1 D2 D3 D4 D5

DQ13
RJK03B7DPA-00/N/7.8m/PPAKSO-8

UGS_1
G
S3 S2 S1

PHS

D1 D2 D3 D4 5

DQ15
RJK0393DPA-0G/N/4.3m/PPAKSO-8

G
S3 S2 S1

LGS_1

DL6

CPU_VAXG
0.68uH/40A/IMD119/M/D

DR167
0/4/SHT/M/X

DR168
0/4/SHT/M/X

CPU_VAXG

DEC14
560u/FP/D/6.3V/69/A/11m

DEC15
560u/FP/D/6.3V/69/A/11m

VIN

DBC82
10u/8/X5R/16V/K

DBC83
10u/8/X5R/16V/K

DBC7
1u/6/X7R/16V/K

DEC12
270u/FP/D/16V/8C/A/10m

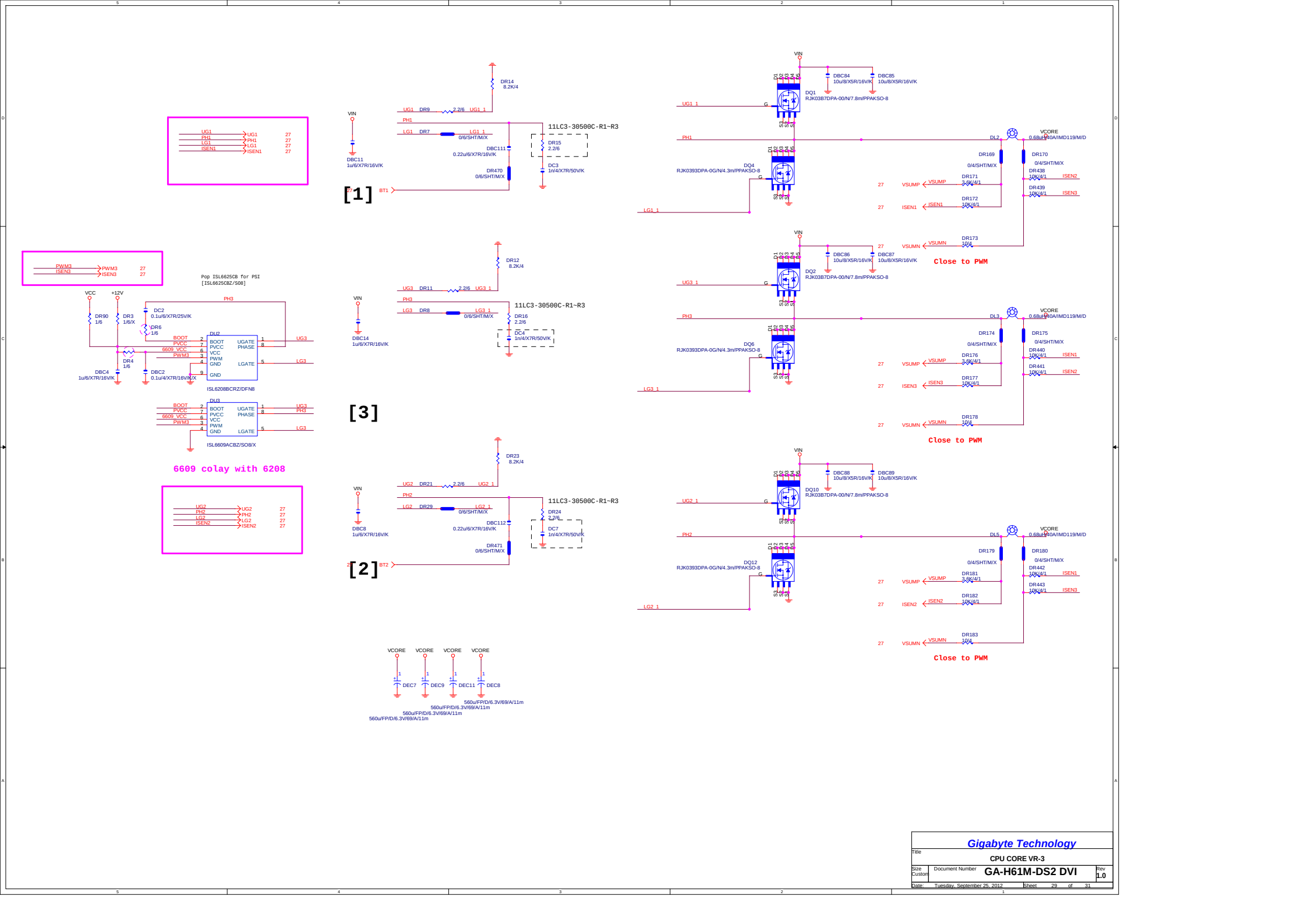
27 VSUMPG VSUMPG
DR472
3.6K/4/1

27 ISEN1G ISEN1G
DR473
10K/4/1/X

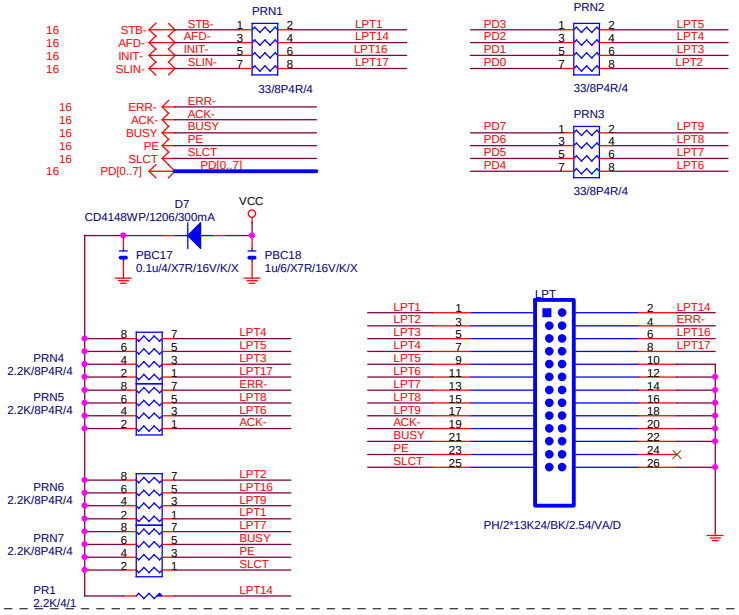
27 VSUMNG VSUMNG
DR474
10/4

Gigabyte Technology

Title	CPU CORE VR-2	
Size: Custom	Document Number: GA-H61M-DS2 DVI	Rev: 1.0
Date: Tuesday, September 25, 2012	Sheet 28 of 31	

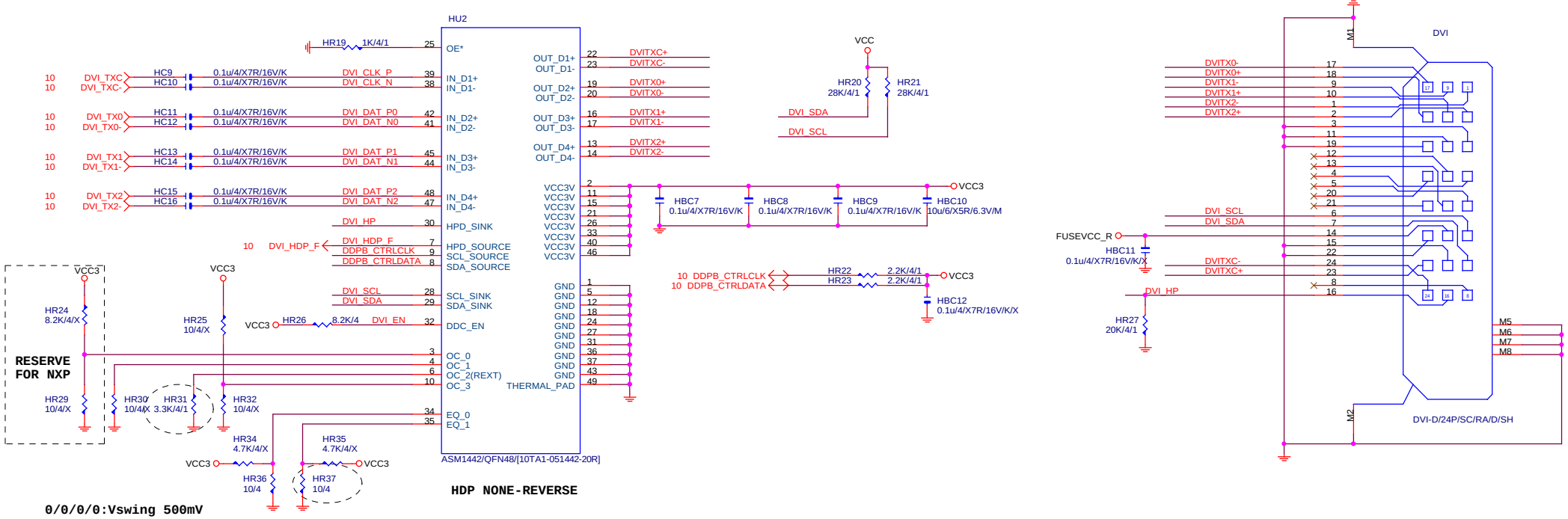

UG1
PH1
LG1
ISEN1
UG1 27
PH1 27
LG1 27
ISEN1 27
PWM3
ISEN3
PWM3 27
ISEN3 27
Pop ISL6625CB for PSI
[ISL6625CBZ/SO8]
VCC
+12V
DR90 1/6
DR3 1/6/X
DR6 1/6
DR4 1/6
DC2 0.1u/6/X7R/25V/K
DBC4 1u/6/X7R/16V/K
DBC2 0.1u/4/X7R/16V/M/X
DU2
BOOT
PVCC
6609_VCC
PWM3
GND
UGATE
PHASE
LGATE
VCC
PWM
ISL6208BCRZ/DFN8
DU3
ISL6609ACBZ/SO8/X
6609 colay with 6208
UG2
PH2
LG2
ISEN2
DR14 8.2K/4
VIN
DBC11 1u/6/X7R/16V/K
UG1_DR9 2.2/6 UG1_1
PH1
LG1_DR7 LG1_1 0/6/SHT/M/X
DBC111 0.22u/6/X7R/16V/K
DR470 0/6/SHT/M/X
11LC3-30500C-R1~R3
DR15 2.2/6
DC3 1n/4/X7R/50V/K
BT1
[1]
DR12 8.2K/4
DBC14 1u/6/X7R/16V/K
UG3_DR11 2.2/6 UG3_1
PH3
LG3_DR8 LG3_1 0/6/SHT/M/X
DR16 2.2/6
DC4 1n/4/X7R/50V/K
[3]
DR23 8.2K/4
DBC8 1u/6/X7R/16V/K
UG2_DR21 2.2/6 UG2_1
PH2
LG2_DR29 LG2_1 0/6/SHT/M/X
DBC112 0.22u/6/X7R/16V/K
DR471 0/6/SHT/M/X
DR24 2.2/6
DC7 1n/4/X7R/50V/K
BT2
[2]
VCORE
DEC7
DEC9
DEC11
DEC8
560u/FP/D/6.3V/69/A/11m
DQ1 RJK03B7DPA-00/N/7.8m/PPAKSO-8
DBC84 10u/8/X5R/16V/K
DBC85 10u/8/X5R/16V/K
UG1_1
PH1
LG1_1
DQ4 RJK0393DPA-0G/N/4.3m/PPAKSO-8
DL2
DR169 0/4/SHT/M/X
DR170 0/4/SHT/M/X
DR171 3.6K/4/1
DR172 10K/4/1
DR438 10K/4/1
DR439 10K/4/1
VSUMP
ISEN1
ISEN2
ISEN3
VSUMN
DR173 10/4
Close to PWM
DQ2 RJK03B7DPA-00/N/7.8m/PPAKSO-8
DBC86 10u/8/X5R/16V/K
DBC87 10u/8/X5R/16V/K
UG3_1
PH3
LG3_1
DQ6 RJK0393DPA-0G/N/4.3m/PPAKSO-8
DL3
DR174 0/4/SHT/M/X
DR175 0/4/SHT/M/X
DR176 3.6K/4/1
DR177 10K/4/1
DR178 10/4
DR440 10K/4/1
DR441 10K/4/1
DQ10 RJK03B7DPA-00/N/7.8m/PPAKSO-8
DBC88 10u/8/X5R/16V/K
DBC89 10u/8/X5R/16V/K
UG2_1
PH2
LG2_1
DQ12 RJK0393DPA-0G/N/4.3m/PPAKSO-8
DL5
DR179 0/4/SHT/M/X
DR180 0/4/SHT/M/X
DR181 3.6K/4/1
DR182 10K/4/1
DR183 10/4
DR442 10K/4/1
DR443 10K/4/1
VCORE 0.68uH/40A/IMD119/M/D
Gigabyte Technology
Title
CPU CORE VR-3
Size Custom
Document Number
GA-H61M-DS2 DVI
Rev 1.0
Date: Tuesday, September 25, 2012
Sheet 29 of 31

LPT PORT



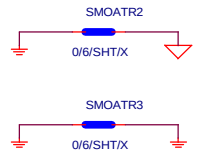
Gigabyte Technology		
Title: **LPT**		
Size: Custom	Document Number: **GA-H61M-DS2 DVI**	Rev: 1.0
Date: Tuesday, September 25, 2012	Sheet 30 of 31	

DVI LEVEL SHIFT



EMI SHORT PAD

PS:視EMI需求



Gigabyte Technology		
Title: DVI		
Size: A3	Document Number: GA-H61M-DS2 DVI	Rev: 1.0
Date: Tuesday, September 25, 2012	Sheet 31 of 31	